UNIVERSITY OF BUEA
FACULTY OF SCIENCE
DEPARTMENT OF PHYSICS

EMBEDDED SYSTEMS: DESIGN AND CONTROL OF A HOME AUTOMATION SYSTEM USING 8051 MICROCONTROLLER AND INTEGRATED CIRCUITS

A project submitted to the Department of Physics in Partial Fulfillment of the requirements for the award of Bachelors of Science (B.Sc.) Degree in Applied Physics

(ELT 498)

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CERTIFICATION
UNIVERSITY OF BUEA
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DEPARTMENT OF PHYSICS

This is to certified that the work described in this project titled:

"EMBEDDED SYSTEMS: DESIGN AND CONTROL OF A HOME AUTOMATION SYSTEM USING 8051 MICROCONTROLLER AND INTEGRATED CIRCUITS"

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Dedication

This work is dedicated to my late parents,

Mr Pevetmi salomon

And

Mrs Nghuebou marie madelaine.

ABSTRACT

The design and control of a home automation system using 8051 microcontroller and ICs interfaced with a Bluetooth module is an innovative and practical solution for improving the quality of life in developing countries. This project aims to provide a user-friendly and efficient system that can control various household appliances such as lights, fans, gates, doorbells, alarms, and security systems. The system uses 8051 microcontroller and ICs to interface with the Bluetooth module and control the appliances through relays. The Bluetooth module allows users to control the appliances remotely using their smartphones or other Bluetooth-enabled devices. The system is designed to be low-cost, easy to install, and maintainable, making it an ideal solution for households in developing countries. The system uses assembly language code to control the appliances, and the simulation is done using software tools such as EdSim51, Proteus, and Keil uVision. The project's success in proving the feasibility and effectiveness of the design and control of a home automation system using 8051 microcontroller and ICs interfaced with a Bluetooth module provides a promising solution for improving the quality of life in developing countries.

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General Introduction

Since the beginning of creation, Human civilization has seen thousands of eras, starting from the early age (stone age) passing through the mechanical age, industrial age and thus our modern era; the computer and electronic age. All such in the aim to achieve a better standard of living with minimum effort. In such, we developed a system known as embedded system which combines our years of research of both software and hardware into a single system which automates our environment using electronics.

In this modern era, life without Electronics is unimaginable [1] and a circuit that allows total control over your equipment without having to move around is a revolutionary concept. Thus, Automation of the surrounding environment of a modern human being allows an increase in his work efficiency and comfort. What an ideal case to study the automation of our environment by starting with the one closest to us, which is our home. [2]

Home automation is a revolutionary technology that has transformed the way we live our lives. It allows us to control various aspects of our homes, such as lighting, temperature, security, and entertainment, using a single system (Embedded system). Home automation is not a new concept in today's world, it is used to provide convenience for users to remotely control and monitor appliances as well as better power management. The efficient use of electricity makes the home automation play an important role in our daily life. [1] There is increasing demand for good homes, where appliances react automatically to changing environmental conditions and might be simply controlled through one common device. Intelligent home automation system is incorporated into smart homes to provide comfort, convenience, and security to home owners. [4] Home automation system represents and reports the status of the connected devices in an intuitive, user-friendly interface allowing the user to interact and control various devices with the touch of a few buttons. Some of the major communication technologies used by today's home automation system include Bluetooth, WiMAX and Wireless LAN (Wi-Fi), Zigbee, and Global System for Mobile Communication (GSM). [3]

The 8051 microcontroller is a popular choice for home automation systems due to its versatility and ease of use. In this essay, we will discuss the basics of home automation using 8051 microcontroller and ICs.

CHAPTER1 : Context and problem statement

1.2 Background of the study

Home automation systems have become increasingly popular in recent years due to the convenience and comfort they provide to homeowners. Thus, they have gained popularity in developed countries due to their ability to enhance the quality of life. These systems allow users to control various aspects of their homes, such as lighting, temperature, security, and entertainment, through a centralized interface. However, existing home automation systems have limitations in terms of functionality, reliability, user-friendliness and thus its adoption in developing countries has been slow due to the high cost and lack of reliable technology. The use of 8051 microcontroller and ICs in the development of home automation systems has been identified as a cost-effective and reliable solution for developing countries. The 8051 microcontroller is a popular choice for embedded systems due to its low cost, ease of programming, and wide availability of support materials. This research aims to develop a home automation system using 8051 microcontroller and ICs that addresses these limitations and will contribute to the development of cost-effective and reliable home automation systems for developing countries, which can improve the living conditions and provide convenience to homeowners. The research will also focus on designing a user-friendly interface that is intuitive and easy to use, and also identify the challenges and opportunities for the adoption of home automation systems in developing countries, which can inform policymakers and stakeholders in the industry.

1.2 Problem statement

The increasing demand for home automation systems has led to the development of various technologies and platforms and thus have become increasingly popular in developed countries, providing homeowners with a convenient and efficient way to control their home appliances and devices. However, there is a need to design an efficient and cost-effective home automation system using 8051 microcontroller and ICs that can cater to the needs of a wide range of households particularly in developing countries where the adoption of home automation systems is still limited due to various challenges such as high costs, lack of technical expertise, and limited availability of resources. Also, the existing systems are either too expensive or lack the necessary features to meet the requirements of modern homes. Additionally, the integration of various devices and sensors into a single platform is a challenging task that requires careful consideration of the hardware and software aspects. Therefore, there is a need for research to develop a comprehensive, cost-effective and reliable home automation system that can be easily installed, configured, controlled by users with minimal technical expertise and can be easily integrated into homes in developing countries. This study aims to address these challenges by designing and developing a functional, reliable and user-friendly prototype of home automation system using 8051 microcontroller and ICs that can provide seamless integration of various devices and sensors for enhanced comfort, security, and energy efficiency in households while evaluating its performance and effectiveness in enhancing the quality of life for homeowners in a developing country.

1.3 Objective and/or Hypothesis

Objective: To design, develop and implement a cost-effective, reliable and user-friendly home automation system using 8051 microcontroller and ICs for a developing country. This system should be able to cater to the needs of modern households, provide seamless integration of various devices and sensors for enhanced comfort, security, and energy efficiency. It should also improve the quality of life of the residents by providing easy control of various home appliances and devices.

Hypothesis: The proposed home automation system using 8051 microcontroller and ICs will be efficient, cost-effective, and user-friendly, providing seamless integration of various devices and sensors for enhanced comfort, security, and energy efficiency in households. The system will be easily installed, configured, and controlled by users with minimal technical expertise, thereby improving the overall quality of life for homeowners in a developing country, leading to a better standard of living and reduced energy consumption.

1.4 Methodology

Methodology:

1. Literature review: Conduct a comprehensive review of literature related to home automation systems, 8051 Microcontroller, and ICs. This will help in understanding the existing systems, their limitations, and the potential benefits of the proposed system.

2. Component selection: Select the appropriate components such as sensors, relays, and switches required for the system. The selection should be based on their compatibility with the 8051 microcontroller and their cost-effectiveness.

3. System design: Based on the literature review and the identified requirements, design a cost-effective home automation system using 8051 microcontroller and ICs. The system should be able to control various home appliances and devices such as lights, fans, air conditioners, alarms, Gates, Television Set and security systems.

4. Circuit design: Design the circuitry for the system using software such as Proteus or Eagle. The circuit should include the 8051 microcontroller, ICs, sensors, relays, switches.

5. Programming: Write the code for the 8051 microcontroller using a programming language such as Assembly or C. The code should be optimized for efficiency and maintenance.

6. Conclusion: Draw conclusions based on the analysis of the data and discuss the

implications of the findings. Provide recommendations for further research and improvements to the system.

1.5 Project Layout

I. Introduction

- Background information on home automation

- Importance of home automation in developing countries

- Objectives of the project

II. Literature Review

- Overview of home automation systems

- Comparison of different microcontrollers and ICs for home automation

III. Methodology

- Selection of components for the project

- Circuit diagram design

- Software program development for the microcontroller

- Hardware assembly and testing

- Mobile application development for remote access

- Performance testing and evaluation

IV. Results and Discussion

- Presentation of the functional home automation system

- Analysis of energy efficiency and cost-effectiveness of the system design

- Evaluation of remote access through the mobile application

- Comparison of the results with existing home automation systems in developing countries

V. Conclusion and Recommendations

- Summary of the research findings
- Implications of the research for developing countries
- Recommendations for future research and implementation of home automation systems

VI. References

- List of sources consulted in the literature review

VII. Appendices

- Circuit diagram and software program code
- Photos and videos of the hardware assembly and testing
- User manual for the mobile application.

CHAPTER 2: Literature review

2.1 INTRODUCTION

The use of home automation systems has become increasingly popular in recent years due to the convenience and efficiency they provide. Home automation systems allow homeowners to remotely control and monitor various household appliances and devices, such as lighting, heating, and security systems. One of the most commonly used microcontrollers for home automation is the 8051 microcontroller. This literature review aims to provide an overview of the research conducted on home automation using 8051 microcontrollers and integrated circuits (ICs). The 8051 microcontroller is a popular choice for home automation due to its low cost, low power consumption, and ease of use. Agarwal (2015) [1] designed a home automation system using the 8051 microcontroller that was able to control various household appliances, such as lights, fans, and air conditioners. The author used a relay module to interface the microcontroller with the appliances, and the system was controlled using a remote control that transmitted signals to the microcontroller via an infrared (IR) receiver.

Chakraborty and Das (2014) [2] also designed a home automation system using the 8051 microcontroller. The system was able to control various household appliances, such as lights, fans, and water pumps. The authors used a relay module to interface the microcontroller with the appliances, and the system was controlled using a keypad that transmitted signals to the microcontroller via a serial communication interface.

Integrated circuits (ICs) are widely used in home automation systems due to their compact size, low power consumption, and high reliability. Kaur and Kaur (2017) [3]designed a home automation system using the 8051 microcontroller and a GSM module. The system was able to control various household appliances, such as lights, fans, and air conditioners. The authors used a temperature sensor and a humidity sensor to monitor the environmental conditions in the house, and the system was controlled using a mobile phone that transmitted signals to the microcontroller via the GSM module.

Kumar and Kumar (2016) [4] designed a home automation system using the 8051 microcontroller and a Bluetooth module. The system was able to control various household appliances, such as lights, fans, and air conditioners. The authors used a motor driver IC to interface the microcontroller with a motorized curtain, and the system was controlled using a smartphone application that transmitted signals to the microcontroller via the Bluetooth module.

Singh and Singh (2015) [5] designed a home automation system using the 8051 microcontroller and a voice recognition module. The system was able to control various household appliances, such as lights, fans, and air conditioners. The authors used a relay driver IC to interface the microcontroller with the appliances, and the system was controlled using voice commands that were recognized by the voice recognition module and transmitted to the microcontroller.

Verma and Kaur (2015) [6] designed a home automation system using the 8051 microcontroller and a Wi-Fi module. The system was able to control various household appliances, such as lights, fans, and air conditioners. The authors used a temperature sensor and a humidity sensor to monitor the environmental conditions in the house, and the system was controlled using a smartphone application that transmitted signals to the microcontroller via the Wi-Fi module.

According to Chong et al. (2016), the 8051 microcontroller is a popular choice for home automation due to its low cost, ease of programming, and ability to interface with various sensors and devices. The authors developed a home automation system that used the 8051 microcontroller to control lighting and temperature, among other features.

In another study, researchers used the 8051 microcontroller in combination with ICs to create a smart home system that could be controlled via an Android application (Khan et al., 2018) [5]. The system allowed users to remotely control various appliances, such as lights and fans, and also provided real-time data on energy consumption.

One of the challenges of home automation using microcontrollers and ICs is ensuring that the system is secure and protected from hackers. To address this issue, researchers have developed various security measures, such as encryption and authentication protocols (Kumar et al., 2019) [6]. These measures can help prevent unauthorized access to the system and protect user privacy.

Another important consideration in home automation is energy efficiency. By using microcontrollers and ICs to control appliances, homeowners can reduce energy consumption and save money on their electricity bills. For example, a study by Al-Ali et al. (2018) [5] found that using a microcontroller-based system to control lighting and air conditioning in a home resulted in significant energy savings. Overall, home automation using microcontrollers and ICs offers many benefits, including convenience, efficiency, and cost savings. However, it is important to ensure that the system is secure and energy-efficient to maximize its potential benefits.

In a review article by A. Kaur et al. (2020) [7], various home automation systems using 8051microcontroller were discussed along with their advantages and limitations. The article highlighted the importance of selecting appropriate ICs for different functionalities such as sensing, control, and communication.

Overall, home automation using 8051 microcontroller and ICs has shown a great potential in improving energy efficiency, convenience, and security in households. However, further research is needed to optimize the design parameters for different applications.

To continue be able to continue with our research and reach our implementation, a brief review of our systems is given bellow:

2.2 EMBEDDED SYSTEMS [8]

As its name suggests, Embedded means something that is attached to another thing. An embedded system can be thought of as a computer hardware system having software embedded in it. An embedded system can be an independent system or it can be a part of a large system. An embedded system is a microcontroller or microprocessor based system which is designed to perform a specific task. For example, a fire alarm is an embedded system; it will sense only smoke. [8]

An embedded system has three components −

- It has hardware.
- It has application software.
- It has Real Time Operating system (RTOS) that supervises the application software and provide mechanism to let the processor run a process as per scheduling by following a plan to control the latencies. RTOS defines the way the system works. It sets the rules during the execution of application program. A small scale embedded system may not have RTOS.

So we can define an embedded system as a Microcontroller based, software driven, reliable, real-time control system. [8]

2.2.1 Characteristics of an Embedded System [8]

- **Single-functioned** − An embedded system usually performs a specialized operation and does the same repeatedly. For example: A pager always functions as a pager.
- **Tightly constrained** − All computing systems have constraints on design metrics, but those on an embedded system can be especially tight. Design metrics is a measure of an implementation's features such as its cost, size, power, and performance. It must be of a size to fit on a single chip, must perform fast enough to process data in real time and consume minimum power to extend battery life.
- **Reactive and Real time** − Many embedded systems must continually react to changes in the system's environment and must compute certain results in real time without any delay. Consider an example of a car cruise controller; it continually monitors and reacts to speed and brake sensors. It must compute acceleration or de-accelerations repeatedly within a limited time; a delayed computation can result in failure to control of the car.
- **Microprocessors based** − It must be microprocessor or microcontroller based.

- **Memory** − It must have a memory, as its software usually embeds in ROM. It does not need any secondary memories in the computer.
- **Connected** − It must have connected peripherals to connect input and output devices.
- **HW-SW systems** − Software is used for more features and flexibility. Hardware is used for performance and security. [8]

Advantages

- Easily Customizable
- Low power consumption
- Low cost
- Enhanced performance

Disadvantages

- High development effort
- Larger time to market

Basic Structure of an Embedded System

The following illustration shows the basic structure of an embedded system −

- **Sensor** − It measures the physical quantity and converts it to an electrical signal which can be read by an observer or by any electronic instrument like an A2D converter. A sensor stores the measured quantity to the memory.
- **A-D Converter** − An analog-to-digital converter converts the analog signal sent by the sensor into a digital signal.
- **Processor & ASICs** − Processors process the data to measure the output and store it to the memory.
- **D-A Converter** − A digital-to-analog converter converts the digital data fed by the processor to analog data
- **Actuator** − An actuator compares the output given by the D-A Converter to the actual (expected) output stored in it and stores the approved output. [8]

2.3 HOME AUTOMATION SYSTEMS/SMART HOMES

2.3.1 Introduction

Home automation involves the use of technology to automate various tasks in a home. The primary goal of home automation is to make life easier and more convenient for homeowners by simplifying routine tasks. Home automation systems can be used to control lighting, heating and cooling systems, security systems, and entertainment systems. On the other hand, a smart home incorporates sensors, actuators, middleware, and a network and has two major interacting components: a smart network and a smart load. The primary objectives of a smart home are to increase home automation, facilitate energy management, and reduce environmental emissions. Home automation can be improved through an improved communication network that involves a twisted pair power lines, radio signals, or fiber optics in a bus-based network or an internet protocol as standards. Smart home energy can be managed through the selection of efficient appliances, improvement in customers' knowledge of and experience with residential energy management, participation in demand-side management (DSM) programs, and the deployment of an energy management system (EMS). Environmental emissions can be decreased through decreased dependency on fossil fuels, the adoption of renewable energy generation sources (such as wind and solar), and the control of electricity consumption.

2.3.2 Home Automation

Although smart devices have been available in the past, their use has been restricted because they lack intercommunication. One suggested solution is to connect all smart devices using hard wiring; however, the resulting portability problem then creates a demand for a wireless network capable of accommodating the devices. Organizations have therefore developed management, additional services, and gateways for smart devices. Smart home network technology has been deployed in different systems, such as power line and radio frequency systems. These developments enable the deployment of a network in smart devices but only with a high associated cost that has itself become a new challenge. A smart home encourages smart living. Quality of life is highly improved through home automation. Home automation reduces energy consumption. Home automation is expensive and hence its speed of evolution is a bit slower. Sooner or later Home Automation will be very popular. Also from the business point of view, it is a flourishing market.

Home automation has been a feature of science fiction writing for many years, but has only become practical since the early 20th Century following the widespread introduction of electricity into the home, and the rapid advancement of information technology. Early remote control devices began to emerge in the late 1800s. For example, Nikola Tesla patented an idea for the remote control of vessels and vehicles in 1898. The emergence of electrical home appliances began between 1915 and 1920[4]; the decline in domestic servants meant that households needed cheap, mechanical replacements. Domestic electricity supply, however, was still in its infancy — meaning this luxury was afforded only the more affluent households. Ideas similar to modern home automation systems originated during the World's Fairs of the 1930s. Fairs in Chicago (1934), New York (1939) and (1964–65), depicted electrified and automated homes. In 1966 Jim Sutherland, an engineer working for Westinghouse Electric, developed a home automation system called "ECHO IV"; this was a private project and never commercialized. The first "wired homes" were built by American hobbyists during the 1960s, but were limited by the technology of the times. The term "smart house" was first coined by the American Association of House builders in 1984. With the invention of the microcontroller, the cost of electronic control fell rapidly. Remote and intelligent control technologies were adopted by the building services industry and appliance manufacturers. By the end of the 1990s,"domotics" was commonly used to describe any system in which informatics and telematics were combined to support activities in the home. The phrase is a portmanteau word formed from domus (Latin, meaning house) and informatics, and refers to the application of computer and robot technologies to domestic appliances. Despite interest in home automation, by the end of the 1990s there was not a widespread uptake, with such systems still considered the domain of hobbyists or the rich. The lack of a single, simplified, protocol and high cost of entry has put off consumers. While there is still much room for growth, according to ABI Research, 1.5 million home automation systems were installed in the US in 2012, and a sharp uptake could see shipments topping over 8 million in 2017.

2.4 PROCESSOR [9]

Processor is the heart of an embedded system. It is the basic unit that takes inputs and produces an output after processing the data. For an embedded system designer, it is necessary to have the knowledge of both microprocessors and microcontrollers. [9]

Processors in a System

A processor has two essential units −

- Program Flow Control Unit (CU)
- Execution Unit (EU)

The CU includes a fetch unit for fetching instructions from the memory. The EU has circuits that implement the instructions pertaining to data transfer operation and data conversion from one form to another.

The EU includes the Arithmetic and Logical Unit (ALU) and also the circuits that execute instructions for a program control task such as interrupt, or jump to another set of instructions.

A processor runs the cycles of fetch and executes the instructions in the same sequence as they are fetched from memory. [9]

2.4.1 Types of Processors

Processors can be of the following categories −

- General Purpose Processor (GPP)
    - Microprocessor
    - Microcontroller
    - Embedded Processor
    - Digital Signal Processor
    - Media Processor
- Application Specific System Processor (ASSP)
- Application Specific Instruction Processors (ASIPs)
- GPP core(s) or ASIP core(s) on either an Application Specific Integrated Circuit (ASIC) or a Very Large Scale Integration (VLSI) circuit.

2.5 Microprocessor

A microprocessor is a single VLSI chip having a CPU. In addition, it may also have other units such as coaches, floating point processing arithmetic unit, and pipelining units that help in faster processing of instructions.

Earlier generation microprocessors' fetch-and-execute cycle was guided by a clock frequency of order of ~1 MHz. Processors now operate at a clock frequency of 2GHz [9]

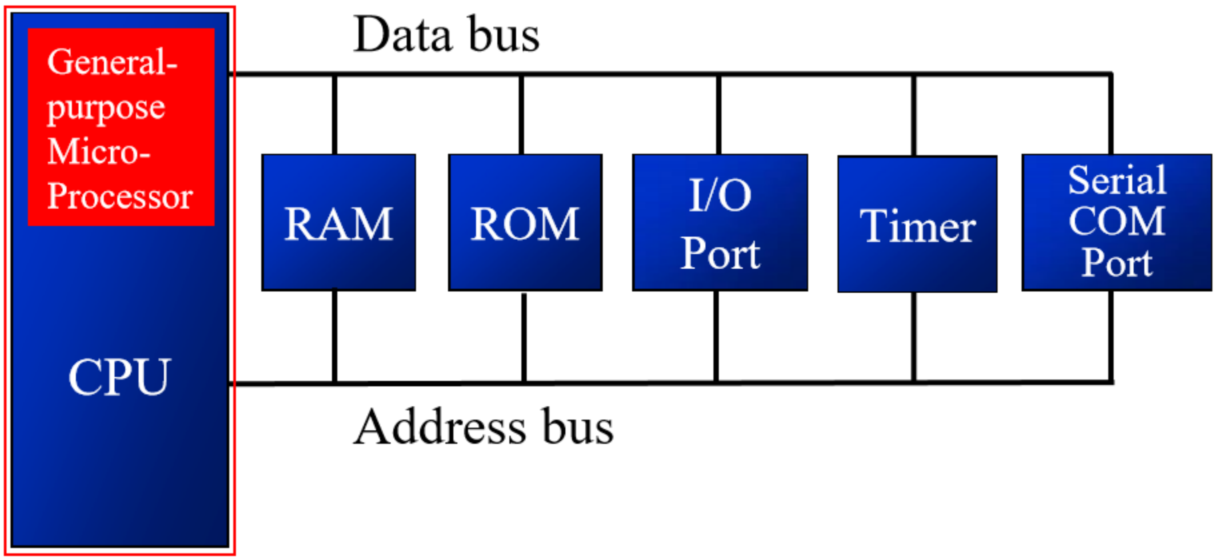


Figure 2.1: Microprocessor Block System

2.5.1 Microcontroller

A microcontroller is a single-chip VLSI unit (also called **microcomputer**) which, although having limited computational capabilities, possesses enhanced input/output capability and a number of on-chip functional units.

CPU	RAM	ROM
I/O Port	Timer	Serial COM Port

Table 2.0: Microcontroller Components

Microcontrollers are particularly used in embedded systems for real-time control applications with on-chip program memory and devices.

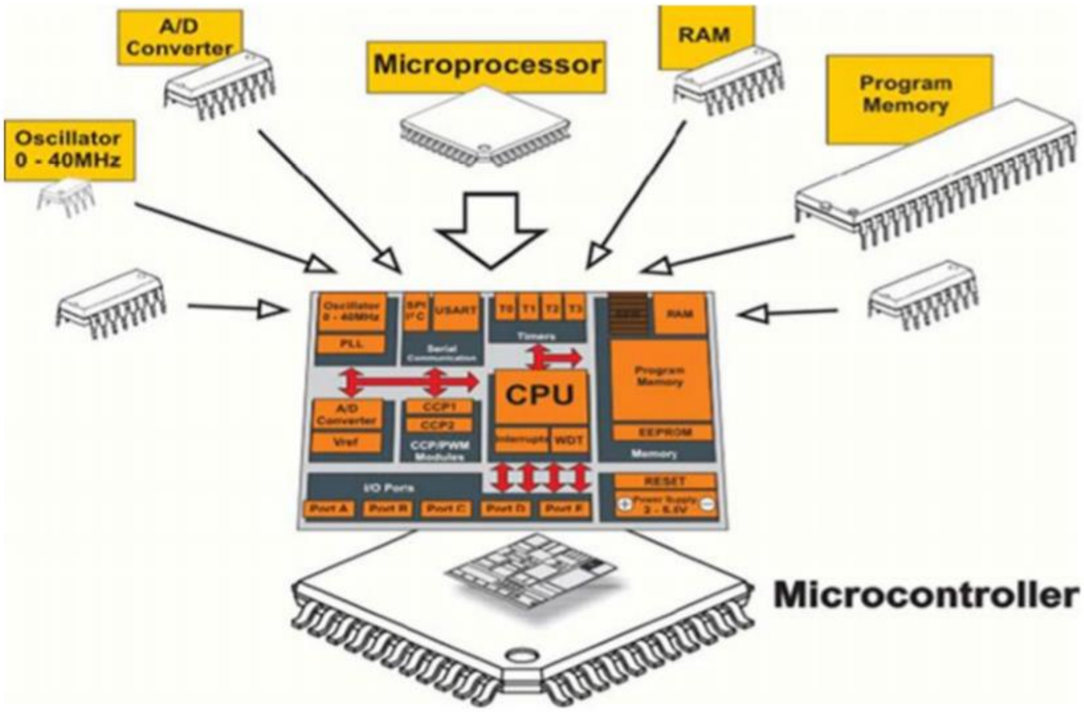


Figure 2.2: Internal Feature of a Microcontroller

2.5.2 Microprocessor vs Microcontroller

Let us now take a look at the most notable differences between a microprocessor and a microcontroller.

Microprocessor	Microcontroller
Microprocessors are multitasking in nature. Can perform multiple tasks at a time. For example, on computer we can play music while writing text in text editor.	Single task oriented. For example, a washing machine is designed for washing clothes only.

RAM, ROM, I/O Ports, and Timers can be added externally and can vary in numbers.	RAM, ROM, I/O Ports, and Timers cannot be added externally. These components are to be embedded together on a chip and are fixed in numbers.
Designers can decide the number of memory or I/O ports needed.	Fixed number for memory or I/O makes a microcontroller ideal for a limited but specific task.
External support of external memory and I/O ports makes a microprocessor-based system heavier and costlier.	Microcontrollers are lightweight and cheaper than a microprocessor.
External devices require more space and their power consumption is higher.	A microcontroller-based system consumes less power and takes less space.

Table 2.2: Differences Between Microcontroller and Microprocessor

2.6 8051 Microcontroller [9]

2.6.1 Brief History of 8051

The first microprocessor **4004** was invented by Intel Corporation. **8085** and **8086** microprocessors were also invented by Intel. In 1981, Intel introduced an 8-bit microcontroller called the **8051**. It was referred as **system on a chip** because it had 128 bytes of RAM, 4K byte of on-chip ROM, two timers, one serial port, and 4 ports (8-bit wide), all on a single chip. When it became widely popular, Intel allowed other manufacturers to make and market different flavors of 8051 with its code compatible with 8051. It means that if you write your program for one flavor of 8051, it will run on other flavors too, regardless of the manufacturer. This has led to several versions with different speeds and amounts of on-chip RAM.

2.6.1 8051 Flavors / Members

- **8052 microcontroller** − 8052 has all the standard features of the 8051 microcontroller as well as an extra 128 bytes of RAM and an extra timer. It also has 8K bytes of on-chip program ROM instead of 4K bytes.
- **8031 microcontroller** − It is another member of the 8051 family. This chip is often referred to as a ROM-less 8051, since it has 0K byte of on-chip ROM. You must add external ROM to it in order to use it, which contains the program to be fetched and executed. This program can be as large as 64K bytes. But in the process of adding external ROM to the 8031, it lost 2 ports out of 4 ports. To solve this problem, we can add an external I/O to the 8031

2.6.2 Comparison between 8051 Family Members

The following table compares the features available in 8051, 8052, and 8031.

Feature	8051	8052	8031
ROM(bytes)	4K	8K	0K
RAM(bytes)	128	256	128
Timers	2	3	2
I/O pins	32	32	32
Serial port	1	1	1
Interrupt sources	6	8	6

Table 2.3: Features of the 8051 Microcontroller

2.6.3 Features of 8051 Microcontroller

An 8051 microcontroller comes bundled with the following features −

- 4KB bytes on-chip program memory (ROM)
- 128 bytes on-chip data memory (RAM)
- Four register banks
- 128 user defined software flags

- 8-bit bidirectional data bus
- 16-bit unidirectional address bus
- 32 general purpose registers each of 8-bit
- 16 bit Timers (usually 2, but may have more or less)
- Three internal and two external Interrupts
- Four 8-bit ports (short model have two 8-bit ports)
- 16-bit program counter and data pointer
- 8051 may also have a number of special features such as UARTs, ADC, Op-amp, etc.

2.6.4 Block Diagram of 8051 Microcontroller

The following illustration shows the block diagram of an 8051 microcontroller –

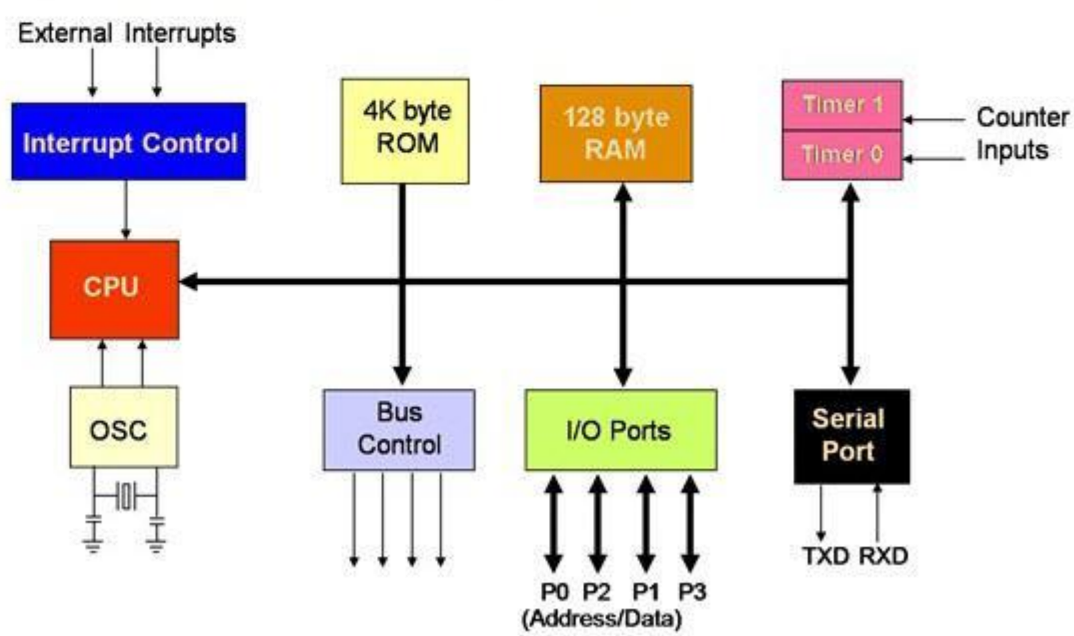


Figure 2.3: The 8051 Block diagram

2.6.5 Pin Description / configuration of the 8051 Microcontroller

8051 family members (e.g., 8751, 89C51, 89C52, DS89C4x0)

– Have 40 pins dedicated for various functions such as I/O, RD, WR, address, data, and interrupts.

– Come in different packages, such as DIP(dual in-line package), QFP(quad flat package), and LLC(leadless chip carrier). Some companies provide a 20-pin version of the 8051 with a reduced number of I/O ports for less demanding applications

- For more information on the pin description, read the following books; 8051
- Microcontroller An application Based Introduction by Chris Braithwaite et Al, [10], The 8051 Microcontroller and embedded System [9]

I/O Port Pins

• The four 8-bit I/O ports P0, P1, P2 and P3 each uses 8 pins.

• All the ports upon RESET are configured as output, ready to be used as input ports by the external device.

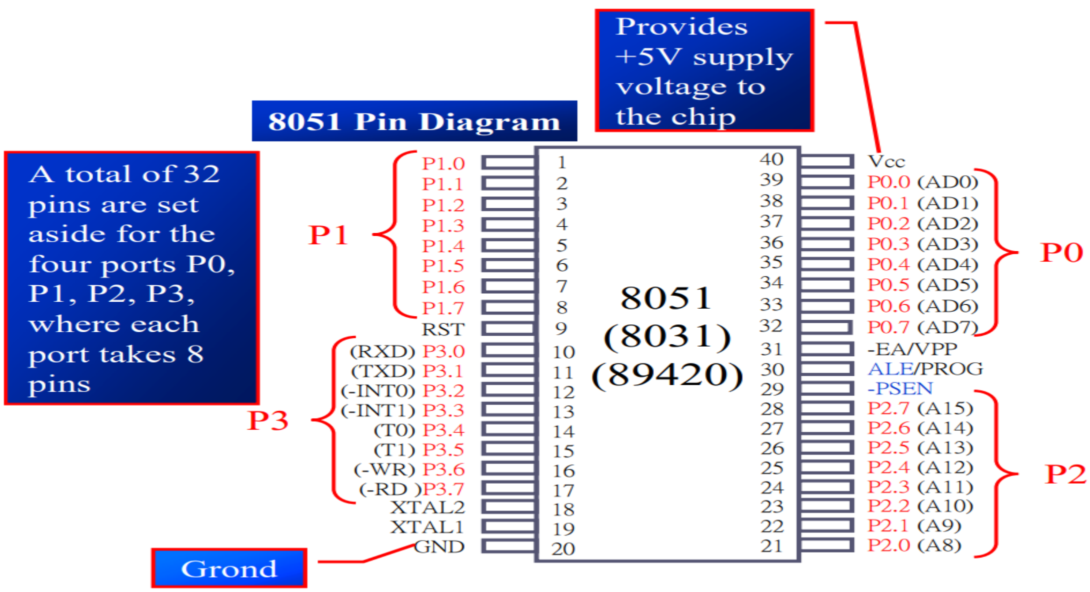


Figure 2.4: 8051 Pin Diagram

Since, this project is not about the deep study of the 8051, only a brief summary of the 8051 microcontroller pin description will be given. The table below summarizes the pin description of the 8051 microcontroller in brief details.

Pin Description Summary

PIN	TYPE	NAME AND FUNCTION
Vss	I	**Ground:** 0 V reference.
Vcc	I	**Power Supply:** This is the power supply voltage for normal, idle, and power-down operation.
P0.0 - P0.7	I/O	**Port 0:** Port 0 is an open-drain, bi-directional I/O port. Port 0 is also the multiplexed low-order address and data bus during accesses to external program and data memory.
P1.0 - P1.7	I/O	**Port 1:** Port I is an 8-bit bi-directional I/O port.
P2.0 - P2.7	I/O	**Port 2:** Port 2 is an 8-bit bidirectional I/O. Port 2 emits the high order address byte during fetches from external program memory and during accesses to external data memory that use 16 bit addresses.
P3.0 - P3.7	I/O	**Port 3:** Port 3 is an 8 bit bidirectional I/O port. Port 3 also serves special features as explained.

Pin Description Summary

PIN	TYPE	NAME AND FUNCTION
RST	I	**Reset:** A high on this pin for two machine cycles while the oscillator is running, resets the device.
ALE	O	**Address Latch Enable:** Output pulse for latching the low byte of the address during an access to external memory.
PSEN*	O	**Program Store Enable:** The read strobe to external program memory. When executing code from the external program memory, PSEN* is activated twice each machine cycle, except that two PSEN* activations are skipped during each access to external data memory.
EA*/VPP	I	**External Access Enable/Programming Supply Voltage:** EA* must be externally held low to enable the device to fetch code from external program memory locations. If EA* Is held high, the device executes from internal program memory. This pin also receives the programming supply voltage Vpp during Flash programming. (applies for 89c5x MCU's)

Figure 2.5: Pin Description Summary of the 8051

2.6.6 8051 Microcontroller Instruction Set [11]

The microcontroller 8051 instructions set includes 110 instructions, 49 of which are single byte instructions, 45 are two bytes instructions and 17 are three bytes instructions. The instructions format consists of a function mnemonic followed by destination and source field. [11]

All the instructions of microcontroller 8051 may be classified based on the functional aspect are given below

- Data transfer group.
- Arithmetic group.
- Logical group.
- Bit manipulation group.
- Branching or Control transfer group.

1.1ADDRESSING MODES

The instructions of 8051 may be classified based on the source or destination type

- Register addressing.
- Direct addressing.
- Register Indirect addressing.
- Immediate addressing.
- Base register + Index register.

All members of the 8051 family execute the same instructions set. The 8051 instructions set is optimized for 8-bit content application. The Intel 8051 has excellent and most powerful instructions set offers possibilities in control area, serial Input/Output, arithmetic, byte and bit manipulation. It has 111 instructions they are

- 49 single byte instructions
- 45 two bytes instructions
- 17 three bytes instructions

The- instructions set is divided into four groups, they are:

- Data transfer instructions

 Arithmetic instructions

 Logical instructions

 Call and Jump instructions

2.1Commonly used instruction of the 8051 microcontroller

Below are some commonly used assembly language instruction by the 8051 microcontroller;

MOV move a byte

SETB set or clear bits

CLR

ACALL call up a subordinate

RET

SJMP unconditional jump

AJMP

JN bit test, conditional jump

JNB

DJNZ byte test, conditional jump

CJNE

ORL OR logic, useful for forcing bits to logic 1

ANL AND logic, useful for forcing bits to logic 0

The full 8051 instruction set can be accessed from the following books with some practical examples;

- 8051 Microcontroller, An Application Based Introduction [10]
- The Art of assembly language [12]
- 8051 assembly language programming/ Instruction Sets [11]
- The 8051 Microcontroller and Embedded Systems [9]

2.7 Conclusion

In conclusion, this chapter, literature review highlights the potential benefits of implementing such systems in homes. In summary, the literature review has demonstrated that home automation systems using 8051 microcontroller and ICs have gained significant attention in recent years due to their potential to enhance the functionality and efficiency of household tasks. The review has highlighted the various components and functionalities of the system, including sensors, actuators, communication protocols, and software platforms. Additionally, the review has identified the challenges and limitations associated with the implementation of the system, such as scalability, reliability, and security concerns. Despite these challenges, the home automation system using 8051 microcontroller and ICs has proven to be a feasible and effective solution for improving energy efficiency, convenience, and security in households. Future research should focus on addressing the technical limitations of the system and developing more robust and scalable solutions to meet the evolving needs of modern households. Overall, the literature review provides valuable insights into the current state of research on home automation systems using 8051 microcontroller and ICs and highlights the potential for further advancements in this field.

CHAPTER 3: Material AND METHODS
(METHODOLOGY)

3.1 Introduction

A system is a combination of components that act together and perform a certain objective. A system is not limited to physical ones. The concept of the system can be applied to abstract, dynamic phenomena such as those encountered in economics. The word system should, therefore, be interpreted to imply physical, biological, economic, and the like, systems. [13].

3.1 Components Selection

To be able to build our home automation system, a careful selection for the required components is needed. This chapter thereby present the various components needed to build our home automation system. Our system will be divided into three parts which are;

- Power system
- Control system
- Output load system

3.2.1 Power System

As in the study of Microcontroller Based Home Automation System with Security, Inderpreet Kaur et Al. [13] stated that; Initial stage of every electronic circuit is power supply system which provides required power to drive the whole system. The specification of power supply depends on the power requirement and this requirement is determined by its rating. The main components used in supply system are:

3.2.3 Transformer

A transformer is an electrical device that transfers energy between two circuits through electromagnetic induction. A transformer may be used as a safe and efficient voltage converter to change the AC voltage at its input to a higher or lower voltage at its output without changing the frequency. Other uses include current conversion, isolation with or without changing voltage and impedance conversion. A transformer most commonly consists of two windings of wire that are wound around a common core to provide tight electromagnetic coupling between the windings. The core material is often a laminated iron core. The coil that receives the electrical input energy is referred to as the primary winding, the output coil is the secondary winding.

Figure 3.1.1: Simple Transformer

The operation of a transformer is based on two principles of the laws of electromagnetic induction: An electric current through a conductor, produces a magnetic field surrounding the conductor, and a changing magnetic field in the vicinity of a conductor induces a voltage across the ends of that conductor. The magnetic field excited in the primary coil gives rise to self-induction as well as mutual induction between coils. This self-induction counters the excited field to such a degree that the resulting current through the primary winding is very small when the secondary winding is not connected to a load.

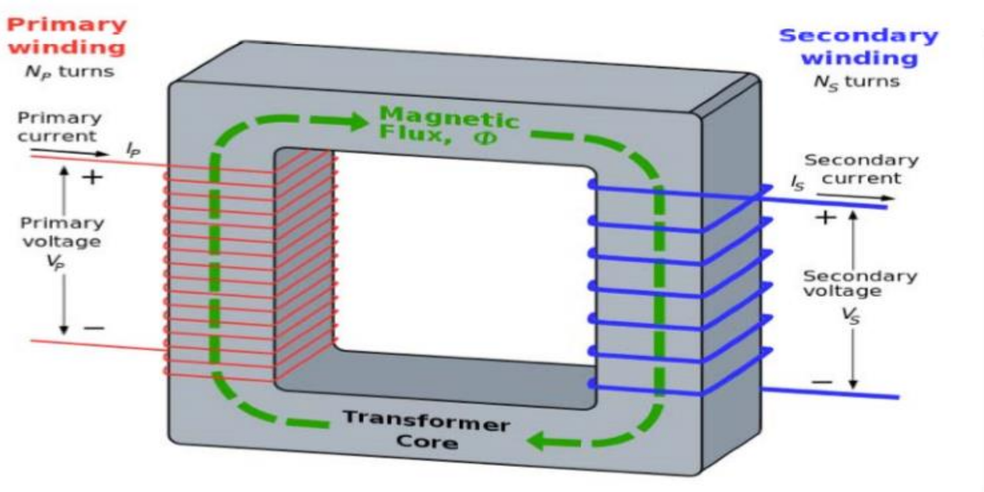
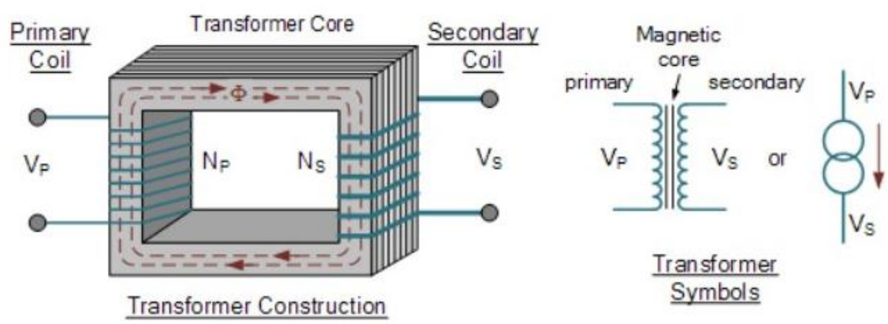


Figure 3.1.2: Transformer Block diagram

A two terminal transformer is to be used to step down main 220V to 12V. It will convert the high voltage from ENEO to a voltage that can be used by the electronic components. The following calculation was considered for the choice of the transformer used for the circuit. The transformer has three terminals both in its input and output of 12Vac – 0Vac – 12Vac of the secondary part. The primary side of the transformer has a rating of 220Vac, while the secondary side has a rating of 12Vdc. [14]



Where:

V_P – is the Primary Voltage

V_S – is the Secondary Voltage

N_P – is the Number of Primary Windings

N_S – is the Number of Secondary Windings

Figure 3.1.3: Transformer construction and symbol

Figure above can operate to either increase or decrease the voltage applied to the primary winding. When a transformer is used to “increase” the voltage on its secondary winding with respect to the primary, it is called a **Step-up transformer**. When it is used to “decrease” the voltage on the secondary winding with respect to the primary it is called a **Step-down transformer** [15].

The relationship between the primary (V_P) and secondary voltage (V_S) is shown in the equation below:

$$V_S = \frac{N_S}{N_P} * V_P$$

If there are more turns on the primary (N_P) than on the secondary (N_S), then the voltage on the secondary will be less than that on the primary. This is a **step-down**

configuration. If the number of turns on the primary is less than the number of turns on the secondary, then the secondary voltage will be higher than that on the primary, resulting in a **step-up configuration**.

Note that the relation $\frac{V_P}{V_S} = \frac{N_P}{N_S} = n$, where n is known as the **turn ratio**. Other relations for the turn ratio are given by;

$\frac{V_P}{V_S} = \frac{N_P}{N_S} = \frac{I_S}{I_P} = n$ and also $(\frac{V_P}{V_S})^2 = \frac{L_P}{L_S}$,

Where L_P is the primary inductance and L_S is the secondary transformer inductance.

3.2.3 *Rectifier*

Rectifier is a circuit which is used to convert ac to dc. Every electronic circuit requires a dc power supply for rectification. There are two types of rectifier circuits which are **half wave and full wave (bridge and bi-phase) rectifier circuits**. For our study, we will make the choice of the bridge full wave rectifier circuit. Bridge rectifier circuit consists of four diodes arranged in the form of a bridge as shown in the figure below. [16]

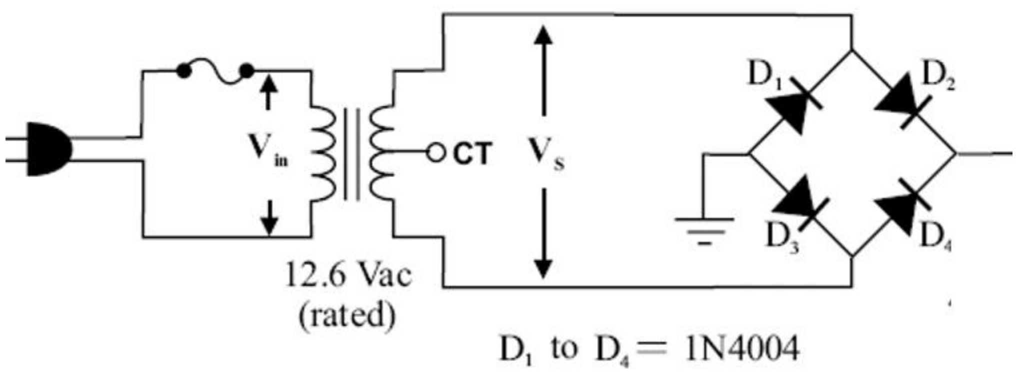


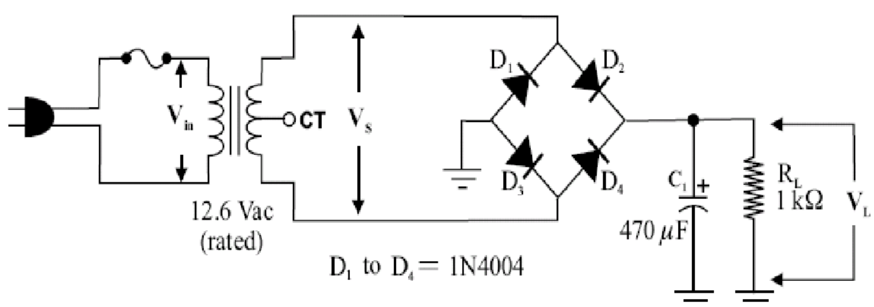
Figure 3.1.4: Unfiltered Full wave bridge rectifier circuit

During the positive half cycle of the input supply, the upper end of the transformer becomes positive with respect to its lower point. This makes one end of the bridge rectifier positive. The diode D1 & D4 become forward biased & D3 & D2 become reverse biased. As a result, current starts flowing from point1, through D1 the load & D4 to the negative end. During negative half cycle, the other end becomes positive. Diodes D1 & D4 now become reverse biased.

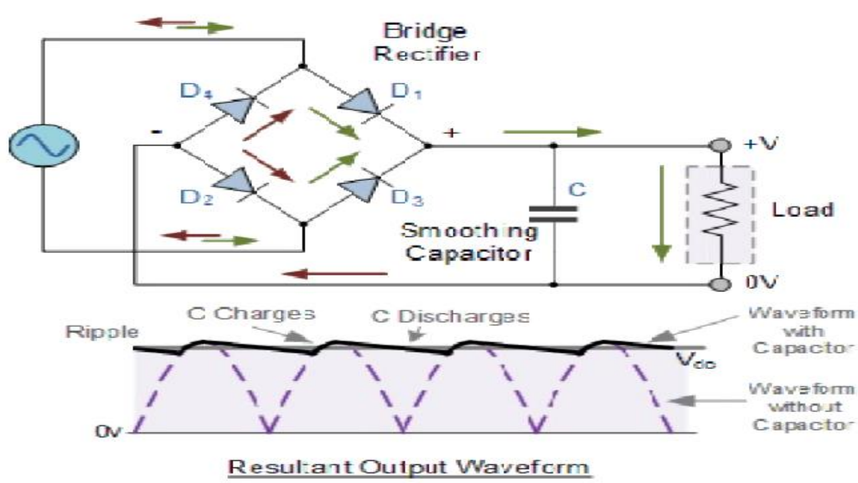
3.2.3 Input filter

After rectification we obtain dc supply from ac but it is not pure dc it may have some ac ripples. To reduce these ripples, we use filters. It comprises of two filters –low frequency ripple filter and high frequency ripple filter. To reduce low frequency ripples we use electrolytic capacitor. The voltage rating of capacitor must be double from incoming dc supply. It blocks dc and passes ripples to ground. The capacitor charges to approximately the peak voltage across the load and then discharges through the load as the rectified DC falls below Vpk. As long as the discharge time for the capacitor is greater than the time between the peaks of the rectified dc, the load voltage can be found using the formula shown. [16]

$$V_{DC} = V_{LPK} - V_r/2$$



a)



b)

Figure 3.1.5: a) filtered full wave rectifier b) output waveform

3.2.3 ***Regulator***

Regulator is a device which provides constant output voltage with varying input voltage. There are two types of regulators-

(a) Fixed voltage regulator

(b) Adjustable regulator

We have used fixed voltage regulator LM78XX last two digits signify output voltage. [13] The 78xx (sometimes L78xx, LM78xx, MC78xx...) is a family of self-contained fixed linear voltage regulator integrated circuits. The 78xx family is commonly used in electronic circuits requiring a regulated power supply due to their ease-of-use and low cost. For ICs within the family, the xx is replaced with two digits, indicating the output voltage (for example, the 7805 has a 5 volt output, while the 7812 produces 12 volts). The 78xx lines are positive voltage regulators: they produce a voltage that is positive relative to a common ground. There is a related line of 79xx devices which are complementary negative voltage regulators. 78xx and 79xx ICs can be used in combination to provide positive and negative supply voltages in the same circuit. [17]

The voltage for our system is 5V that is why we have used 7805 regulator which provides 5V from 12V dc.

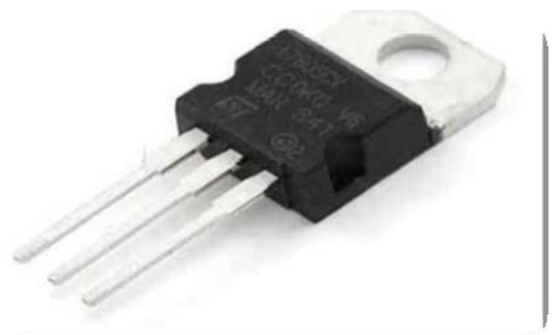


Figure 3.1.6 LM78XX Voltage regulator

78xx ICs have three terminals and are commonly found in the TO220 form factor, although smaller surface-mount and larger TO3packages are available. These devices support an input voltage anywhere from a few volts over the intended output voltage, up to a maximum of 35 to 40 volts depending on the make, and typically provide 1 or 1.5 amperes of current (though smaller or larger packages may have a lower or higher current rating).

Part Number	Output Voltage
7805	+5 V
7812	+12 V

Table 3.1: voltage specification for LM7805 and LM7812

3.2.3 *Output filter*

It is used to filter out output ripple if any. And it makes use of a capacitor. A capacitor (originally known as a condenser) is a passive two-terminal electrical component used to store energy electrostatically in an electric field. The forms of practical capacitors vary widely, but all contain at least two electrical conductors (plates) separated by a dielectric (i.e., insulator). The conductors can be thin films of metal, aluminum foil or disks, etc. The 'non-conducting' dielectric acts to increase the capacitor's charge capacity. A dielectric can be glass, ceramic, plastic film, air, paper, mica, etc. Capacitors are widely used as parts of electrical circuits in many common electrical devices. Unlike a resistor, a capacitor does not dissipate energy. Instead, a capacitor stores energy in the form of an electrostatic field between its plates.

Figure 3.1.7: Capacitors

Capacitors are widely used in electronic circuits for blocking direct current while allowing alternating current to pass.

In analog filter networks, they smooth the output of power supplies.

In resonant circuits they tune radios to particular frequencies.

In electric power transmission system, they stabilize voltage and power flow.

3.2.3 Output indication

We use LED to observe the functioning of our system. If the LED glows it confirms proper functioning of our supply. We have used four power supply units. A light-emitting diode (LED) is a two-lead semiconductor light source that resembles a basic pn-junction diode, except that an LED also emits light.

When an LED's anode lead has a voltage that is more positive than its cathode lead by at least the LED's forward voltage drop, current flows.

Electrons are able to recombine with holes within the device, releasing energy in the form of photons. This effect is called electroluminescence, and the color of the light (corresponding to the energy of the photon) is determined by the energy band gap of the semiconductor. A LED is often small in area (less than 1 mm2), and integrated optical components may be used to shape its radiation pattern

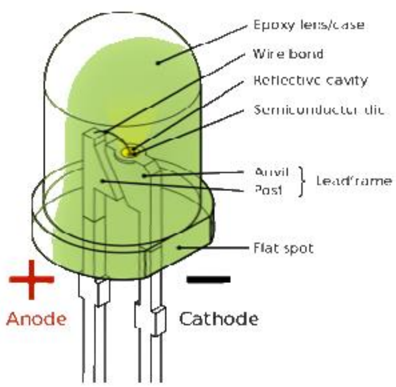


Figure 3.1.8: LED’s block diagram and some color types

3.2.3 Control System

A control system for a project on home automation using the 8051 microcontroller and ICs can be designed as follows:

1. Microcontroller: The 8051 microcontroller will be the brain of the system. It will receive inputs from various sensors and switches, and based on the programmed logic, it will control the various devices in the home.

A brief overview has been given in chapter 2 alongside with its basic instruction set.

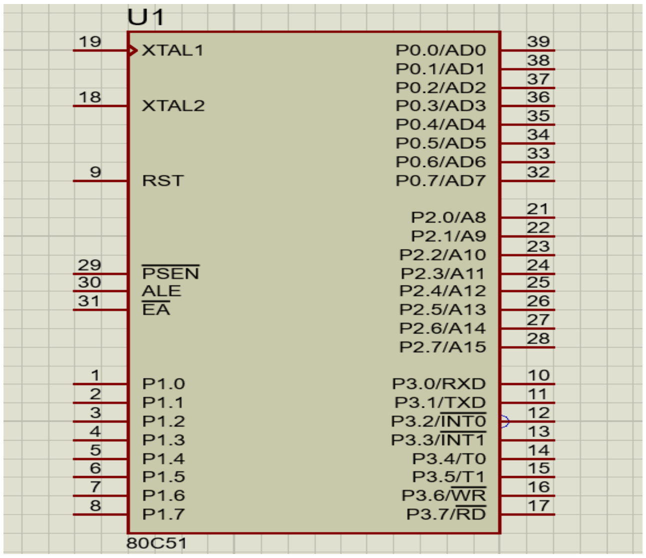


Figure 3.2.1: 8051 microcontroller pin diagram on Proteus

1. Bluetooth Module

HC-05 Bluetooth module consists two things one is Bluetooth serial interface module and a Bluetooth adaptor. Bluetooth serial module is used for converting serial port to Bluetooth. The HC-05 may be a terribly cool module which may add two-way (full-duplex) wireless functionality in our projects. We can use this module to communicate between microcontrollers with any device with Bluetooth functionality like a tablet, phone or laptop. There are many android & IOS applications that are already available which made the process easier. The module communicates with the assistance of USART at 9600 baud thus it's simple to interface with any microcontrollers which supports USART. We are able to additionally set up the default values of the module by using the command mode. So, if you looking for a Wireless module that would transfer Character from your computer or mobile phone to microcontroller. And this is the best choice for us. But this module cannot transfer multimedia like, songs or photos; so for transferring the data we choose the HC-05 interface. [18] The HC-06 works with a supply voltage of 3.6 VDC to 6 VDC, however, the logic level of RXD pin is 3.3 V and is not 5 V tolerant. [19]

Figure3.2.2: HC-05 Bluetooth module front and back view

The process of connecting the 8051 Microcontroller and the Bluetooth module follows the steps 1–3 to enable the android application to control the 8051 (microcontroller). [20]

Step 1: The Bluetooth module and microcontroller was connected through their respective VCC pins.

Step 2: The Bluetooth module and microcontroller was connected through their respective GND pins.

Step 3: Similarly, the transmitter (Tx pin) of the Bluetooth module to the receiver (Rx pin) of the microcontroller and the receiver (Rx pin) of the Bluetooth module to the transmitter (Tx pin) of the microcontroller.

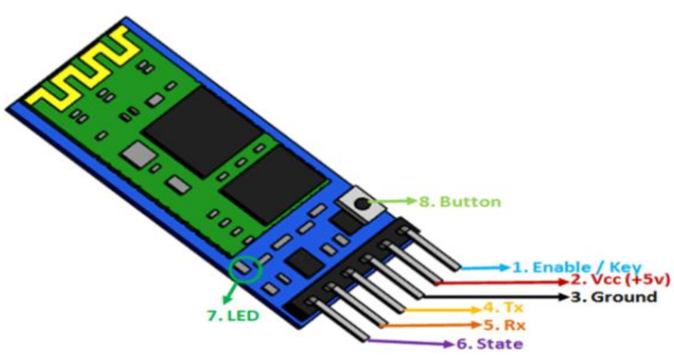


Figure3.2.3: HC-05 Bluetooth configuration pins

3.Android Mobile Device

Android is an Operating System for smart phone devices on which we can run our application. Android provides healthy array of connectivity options including Bluetooth and wireless data over a cellular connection. Android provides access to a wide range of useful libraries and tools that can be used to build rich applications.

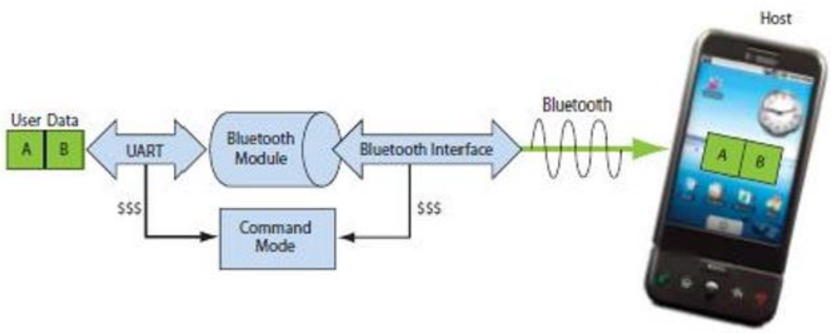


Figure 3.2.4: Block diagram of transfer of data from Bluetooth module to phone

To run this project, first we need to download Bluetooth app form Google play store. We can use any Bluetooth app that can send data using Bluetooth. Here are some apps name that can be used:

1. Bluetooth Spp pro
2. Bluetooth controller

Figure3.2.5: Bluetooth app interface

4.Relays: Relays will be used to control high voltage devices such as air conditioners, geysers, and water pumps. [21] A relay is an electrically operated switch. Many relays use an electromagnet to mechanically operate a switch, but other operating principles are also used, such as solid-state relays. Relays are used where it is necessary to control a circuit by a separate low-power signal, or where several circuits must be controlled by one signal. The first relays were used in long distance telegraph circuits as amplifiers: they repeated the signal coming in from one circuit and re-transmitted it on another circuit. Relays were used extensively in telephone exchanges and early computers to perform logical operations. A type of relay that can handle the high power required to directly control an electric motor or other loads is called a contactor. The Relay input voltage is 5v, so the relay output 10 amps / 250v A.C and the signal input 5v D.C from microcontroller.

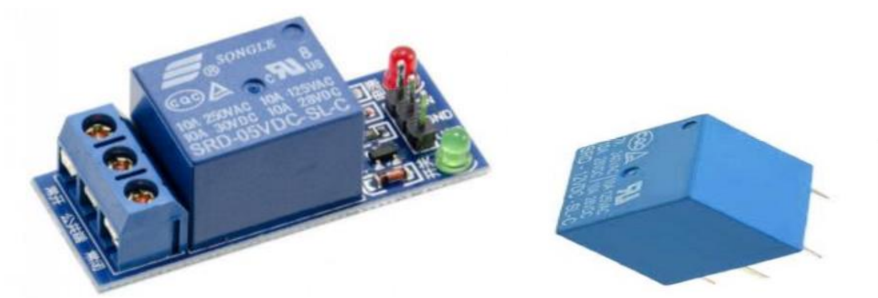


Figure3.2.6: Relay Module

Types of relays [22]

- Electromagnetic Relays: These relays are constructed from electrical, mechanical and magnetic components, and possess operating coil and mechanical contacts. Therefore, when a coil gets activated by a power supply source, these mechanical contacts get opened or closed. The type of supply can be AC or DC.

- Solid State or Electronic Relays: Solid State uses solid state components to perform the switching operation with one or more semiconductor switching devices like a power transistor, thyristor and TRIAC without moving any parts. Since the control energy required is much lower, compared to the output power to be controlled by this relay. [23]

- High Voltage Relays: These relays are quite similar in function to that of low voltage relays, but the major difference is the contacts which are designed to operate at high voltages. Therefore, a high insulation is provided between the contacts.

 Time Delay Relays: The time-delay relays are used for performing time delayed switching operations such as starting, protecting and controlling circuits applications.

 Thermal Relay: These relays are based on the effects of heat, which means – the rise in the ambient temperature from the limit, directs the contacts to switch from one position to another.

Some applications of Relays:

 Control a high-voltage circuit with a low-voltage signal, as in some types of modems or audio amplifiers.

 Control a high-current circuit with a low-current signal, as in the starter solenoid of an automobile.

 Detect and isolate faults on transmission and distribution lines by opening and closing the circuit breakers.

 Time delay functions. Relays can be modified to delay opening or delay closing set of contacts. A very-short delay uses a copper disk between the armature and moving blade assembly.

5. Relay Module:

Since one of our project was to maximize energy efficiency, the use of the IC ULN 2003A is more appropriate to use than to use a transistor for each device that needs to be interfaced.

The relay driver circuit is powered by Darlington IC ULN2003A that has seven pairs of a transistor with seven inputs and output pins. The Darlington IC used eliminates the use of multiple components. The relay is powered by 12 V from the power supply and it has two terminals, one terminal is connected to the 12 V supply while the other terminal is connected to 5 V from the 8051 microcontroller which acts like a neutral that activates the relays. It can drive many of Applications include relay drivers, lamp drivers, hammer drivers, display drivers, line drivers, and logic buffers. [18] The Darlington IC has the circuitry inside it that can increase the current to the relays and it has seven transistorized output pins, which eliminate the use of several single transistors, but the output is only activated when the input pin of the IC is supplied with lower voltage. The Darlington IC ULN2003A is shown below;

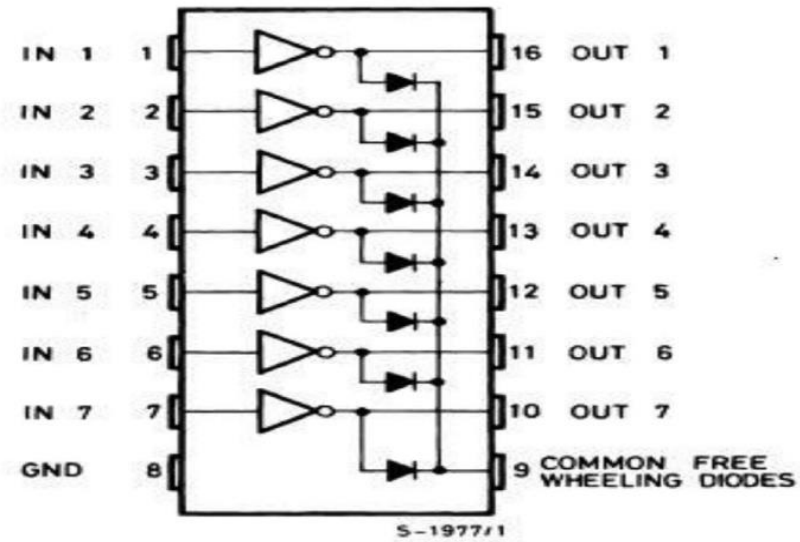


Figure 3.2.7: Darlington IC (ULN2003A)

6. LCD display: An LCD display will be used to display the status of the devices in the home. Display units are the most important output devices in embedded projects and electronics products. 16x2 LCD is one of the most used display unit. 16x2 LCD means that there are two rows in which 16 characters can be displayed per line, and each character takes 5X7 matrix space on LCD. In this tutorial we are going to connect 16X2 LCD module to the 8051 microcontroller (AT89S52). Interfacing LCD with 8051 microcontroller might look quite complex to newbies, but after understanding the concept it would look very simple and easy. Although it may be time taking because you need to understand and connect 16 pins of LCD to the microcontroller. So first let's understand the 16 pins of LCD module.

Figure 3.2.8: LCD Display

We can divide it in five categories, Power Pins, contrast pin, Control Pins, Data pins and Backlight pins.

Category	**Pin NO.**	**Pin Name**	**Function**
Power Pins	1	VSS	Ground Pin, connected to Ground
	2	VDD or Vcc	Voltage Pin +5V
Contrast Pin	3	V0 or VEE	Contrast Setting, connected to Vcc thorough a variable resistor.
Control Pins	4	RS	Register Select Pin, RS=0 Command mode, RS=1 Data mode
	5	RW	Read/ Write pin, RW=0 Write mode, RW=1 Read mode
	6	E	Enable, a high to low pulse need to enable the LCD
Data Pins	7-14	D0-D7	Data Pins, Stores the Data to be displayed on LCD or the command instructions
Backlight Pins	15	LED+ or A	To power the Backlight +5V
	16	LED- or K	Backlight Ground

Table 3.2: LCD specifications

All the pins are clearly understandable by their name and functions, except the control pins, so they are explained below:

RS: RS is the register select pin. We need to set it to 1, if we are sending some data to be displayed on LCD. And we will set it to 0 if we are sending some command instruction like clear the screen (hex code 01).

RW: This is Read/write pin, we will set it to 0, if we are going to write some data on LCD. And set it to 1, if we are reading from LCD module. Generally, this is set to 0, because we do not have need to read data from LCD. Only one instruction "Get LCD status", need to be read some times.

E: This pin is used to enable the module when a high to low pulse is given to it. A pulse of 450 ns should be given. That transition from HIGH to LOW makes the module ENABLE.

There are some preset command instructions in LCD, we have used them in our program below to prepare the LCD (in lcd_init() function). Some important command instructions are given below:

Hex Code	**Command to LCD Instruction Register**
0F	LCD ON, cursor ON
01	Clear display screen
02	Return home
04	Decrement cursor (shift cursor to left)
06	Increment cursor (shift cursor to right)
05	Shift display right

07	Shift display left
0E	Display ON, cursor blinking
80	Force cursor to beginning of first line
C0	Force cursor to beginning of second line
38	2 lines and 5×7 matrix
83	Cursor line 1 position 3
3C	Activate second line
08	Display OFF, cursor OFF
C1	Jump to second line, position 1
OC	Display ON, cursor OFF
C1	Jump to second line, position 1
C2	Jump to second line, position 2

Table 3.3: Some LCD command instructions

3.2.4 Output Loads

1.Electric light

Like in the saying, “In the beginning God said: let there be light and there was light”, [24]and without light we cannot see, the first component of our home automation system, will be bringing light to our homes. And for us to do that, we must first understand what light is.

The electric light, one of the everyday conveniences that most affects our lives, was not “invented” in the traditional sense in eighteen seventy-nine by Thomas Alva Edison, although he could be said to have created the first commercially practical incandescent light. He was neither the first nor the only person trying to invent an incandescent light bulb. [25]

Types of Lights

 Indoor Lighting: Some of the most common indoor light bulbs are incandescent bulbs, which look like a traditional light bulb. Generally, the input for these bulbs is either 40W or 60W. But there are other kinds of indoor light bulbs as well.

 Incandescent Bulbs: The incandescent light bulb has had the same design for over 100 years since Thomas Edison invented it! It produces light when a thin wire called a tungsten filament is heated by electricity running through it making it so hot that it starts to glow brightly.

 Compact Fluorescent Light Bulbs (CFL): These spiraled light bulbs are far more efficient than the standard incandescent bulb. Compact Fluorescent Light bulbs (CFLs) work by running electricity through gas inside the coils, exciting that gas, and producing light.

 Light Emitting Diode (LED): Unlike incandescent and CFL bulbs, LED bulbs have moved into the technological age. LEDs that produce white light work in a rather complicated way.

 Outdoor Lighting: Outdoor lights are usually different from those bulbs used indoors because they need to be much brighter and last longer.

 Halogen Bulbs Halogen bulbs are often found in homes as spotlights or floodlights.

 Metal Halide: Metal halide lamps are commonly used in streetlights.

 High Pressure Sodium (HPS): The high pressure sodium lamp (HPS) is the most commonly used street light throughout the world. It produces light by running electricity through a mixture of gases.

2.Fans

Any device that produces a current of air by the movement of broad surfaces can be called a fan. Fans fall under the general classification of "turbo machinery" Fans fall under the general classification of turbo machinery and have a rotating impeller at least partially encased in a stationary housing. Fans are similar in many respects to pumps. Both are turbo machines that transfer energy to a flowing fluid [26].

Fan Types

Fans are classified according to the direction of flow through the impeller:

 Axial Flow: Air flows through the impeller parallel to, and at a constant distance from the axis. The pressure rise is provided by the direct action of the blades

 Centrifugal or radial flow: Air enters parallel to the axis of the fan and turns through 900and is discharged radically through the blades. The blade force is tangential causing the air to spin with the blades and the main pressure is attributed to this centrifugal force.

 Mixed flow: Air enters parallel to the axis of the fan and turns through an angle which may range from 300 to 900. The pressure rise is partially by direct blade action and partially by centrifugal action.

 Cross Flow: Air enters the impeller at one part of the outer periphery flows inward and exits at another part of the outer periphery

3.Switching devices

In electrical engineering, a switch is an electrical component that can disconnect or connect the conducting path in an electrical circuit, interrupting the electric current or diverting it from one conductor to another.

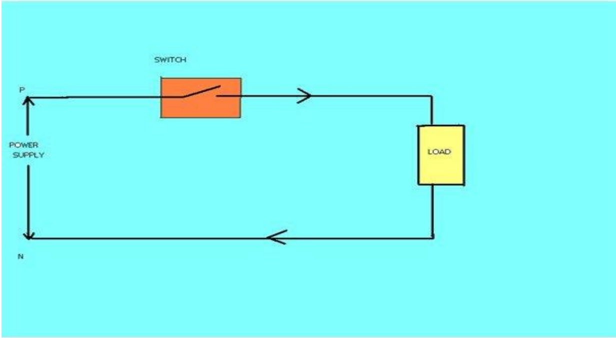


Figure3.3.1: Switching a load

In our home automation system, many devices will work as loads requiring switching operations. Some of these devices include;

a.TV Set

A TV set will also be considered as a load and will be controlled on the operation of a switch.

b. A Gate

since a gate is a component that functions on the principle of open and closed, thus a switching system is required for its operation.

c. An alarm or doorbells

From first thought, it seems like both the doorbell and alarm don't work on a switch, but they do. They both work on the principle of a buzzer. A buzzer in a home automation system is a device that produces an audible sound to alert the homeowner of an event or trigger. It can be used to indicate the opening or closing of a door, the activation of a security system, or the completion of a task. The buzzer can be connected to the home automation controller and programmed to produce different sounds and tones depending on the event or trigger. It is an essential component of a comprehensive home automation system that provides alerts and notifications to keep the homeowner informed and aware of what is happening in their home.

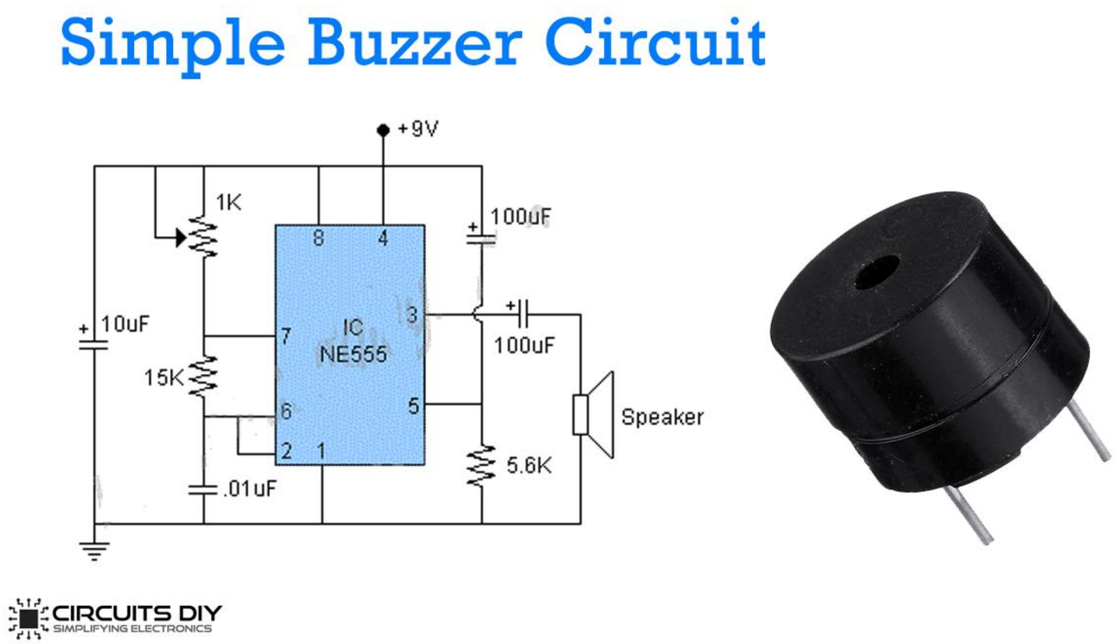


Figure 3.3.2: Internal diagram of a simple buzzer

3.3 SYSTEM DESIGN

This section is concerned with the design of our system. Based on the literature review and the identified requirements, we will design a cost-effective home automation system using 8051 microcontroller and ICs. The system will be able to control various home appliances and devices such as lights, fans, air conditioners, and security systems.

To be able effectively design our system, I made use of some software;

- Proteus

Proteus is a software tool used for simulation and design of electronic circuits. It is widely used by engineers, students, and hobbyists for designing and testing electronic circuits before they are physically built. Proteus provides a virtual environment where users can design, simulate, and test their circuits without the need for physical components. It includes a wide range of components such as microcontrollers, sensors, and actuators, and allows users to create custom components as well. Proteus also includes a powerful simulation engine that can simulate complex circuits in real-time, allowing users to test their designs and troubleshoot any issues before building the circuit. Additionally, Proteus includes a PCB design tool that allows users to design and layout their circuits for manufacturing. Overall, Proteus is a powerful tool for electronic circuit design and simulation, and is widely used in the industry and academia.

- Keil u Vision

Keil uVision is an integrated development environment (IDE) used for developing software for microcontrollers. It is widely used by embedded systems engineers for developing firmware for microcontrollers such as the 8051, ARM, and Cortex-M series. Keil uVision provides a comprehensive set of tools for developing, debugging, and testing embedded software. It includes a code editor, compiler, linker, and debugger, as well as a range of software libraries and example code. Keil uVision also includes a simulator that allows users to test their code without the need for physical hardware. Additionally, Keil uVision supports a wide range of microcontrollers and provides a user-friendly interface for configuring and programming them. Overall, Keil uVision is a powerful tool for developing embedded software and is widely used in the industry and academia

- EDSIM51

. EdSim51 is a simulator for the 8051 microcontroller that allows users to write, assemble, and simulate 8051 assembly language programs. It provides a graphical user interface that allows users to interact with the simulated microcontroller and its peripherals, such as LEDs, switches, and LCD displays. EdSim51 is a useful tool for learning and practicing 8051 assembly language programming, as well as for testing and debugging 8051-based projects before deploying them on actual hardware.

To achieve our goal, our system will be divided into three sections as explained in the section 3.1 in this chapter. The sections are as follows:

- Power system
- Control system
- Output load system

To view that clearly, below is a schematic for our system.

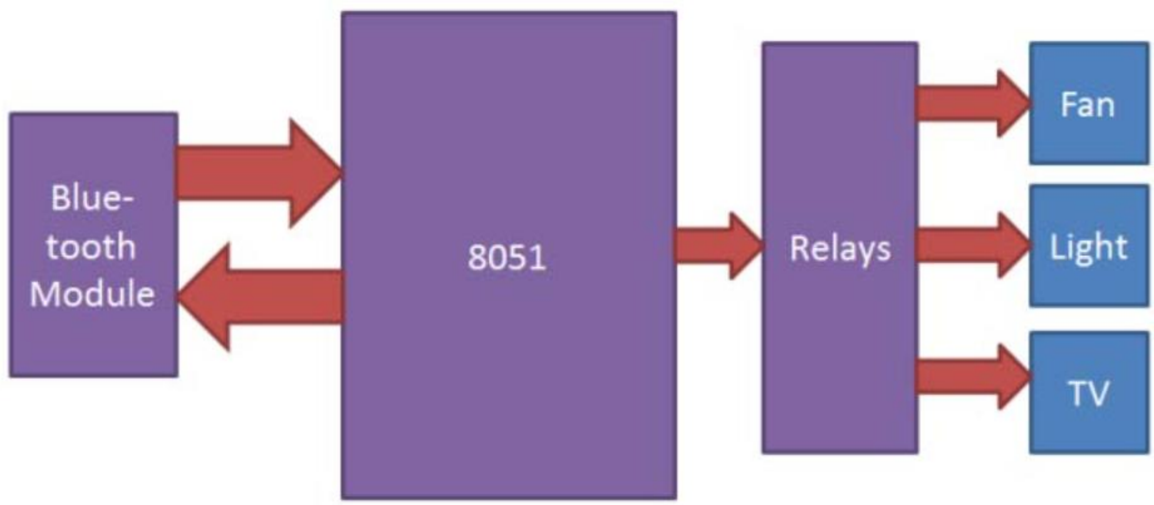


Figure 3.4.0: Home Automation System Block diagram.

In the schematic above, the Bluetooth module is our input part of the system, the 8051 Microcontroller the control system part and the relays and components like Fan, Light, TV, etc., the output load system part. In the system above, the system is said to be already supplied by the required power. Thus, Power system is not included above. All the necessary components required for our system have already been discussed in section 3.1. as stated by our research topic and objectives, we shall proceed with the design of home automation using 8051 microcontroller and Integrated Circuits satisfying the following conditions:

- Manage and Reduce energy consumption thereby, increasing energy efficiency.
- Cost-effective and user-friendly.
- The system will be easily installed, configured, and controlled by users with minimal technical expertise.
- Provide seamless integration of various devices and sensors for enhanced comfort, security and improve the quality of life of the residents by providing easy control of various home appliances and devices.

To get a better understanding on our system, the following block diagram shows the operation of whole system which is going to operate by the following manner.

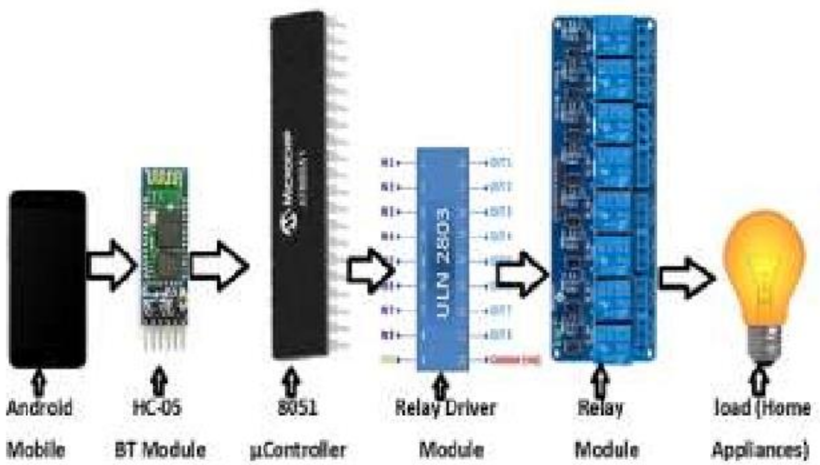


Figure 3.4.1: Home automation system operation

- Android mobile
- Bluetooth module (HC-05)
- 8051 µController (AT89S51)
- Relay driver module (ULN2803A)
- Relay module (8 channel)
- Home appliances (fan, motor, light, etc.)

3.3.1 Power system

To be able to achieve our first two objectives, I built a power system on Proteus which is both power effective and cost effective. Our power is to provide a 5V for our microcontroller and 12V for the relays. Below is the diagram on Proteus providing 5V DC using LM7805 IC and 12V using LM7812.

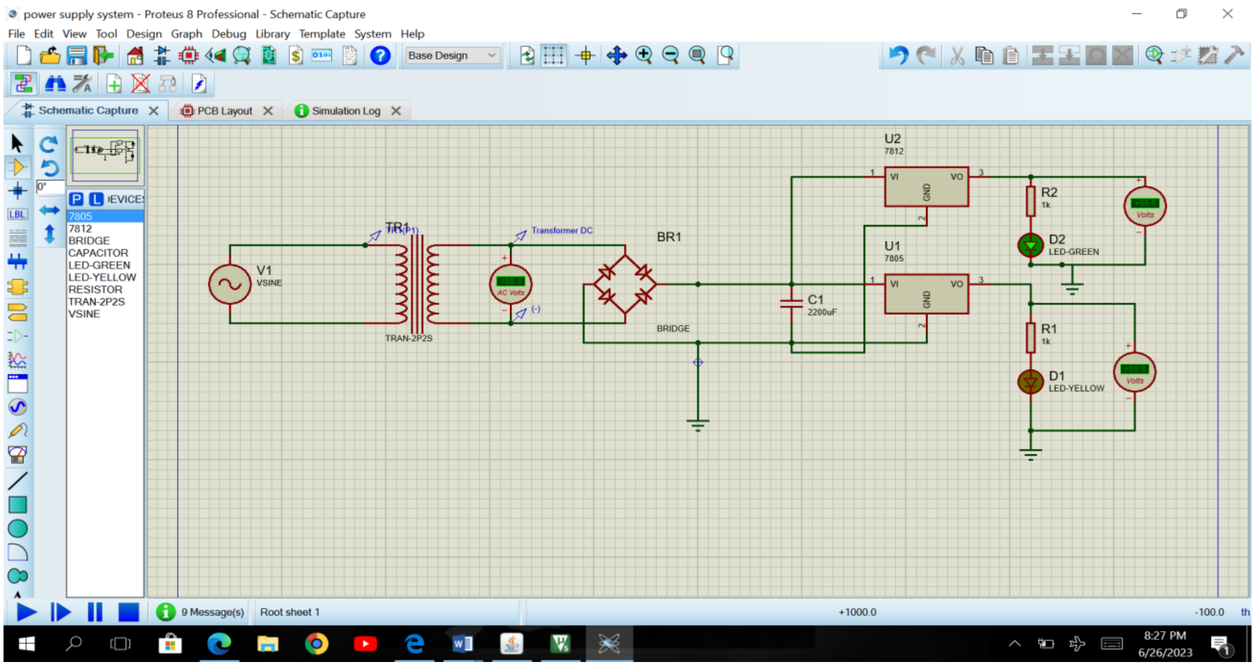


Figure 3.4.2: Power control circuit

For our power supply, we had an AC signal input 0f 220V with a transformer ratio of 1:5.83, we obtained a secondary voltage of 12 V. Below is the graph of the AC primary and secondary voltage.

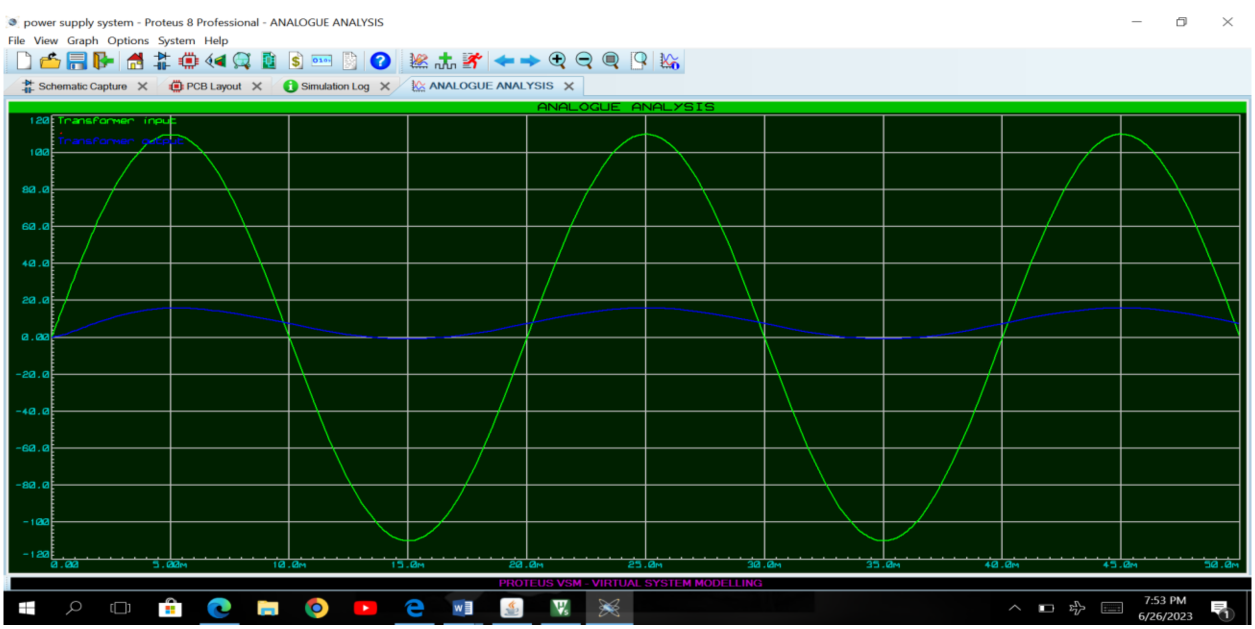


Figure 3.4.4: A.C voltage input wave

At the level of our loads after rectification, I obtained a 5V and 12V for LM7805 and LM7812 respectively. The graph below displays the DC output for our two ICs.

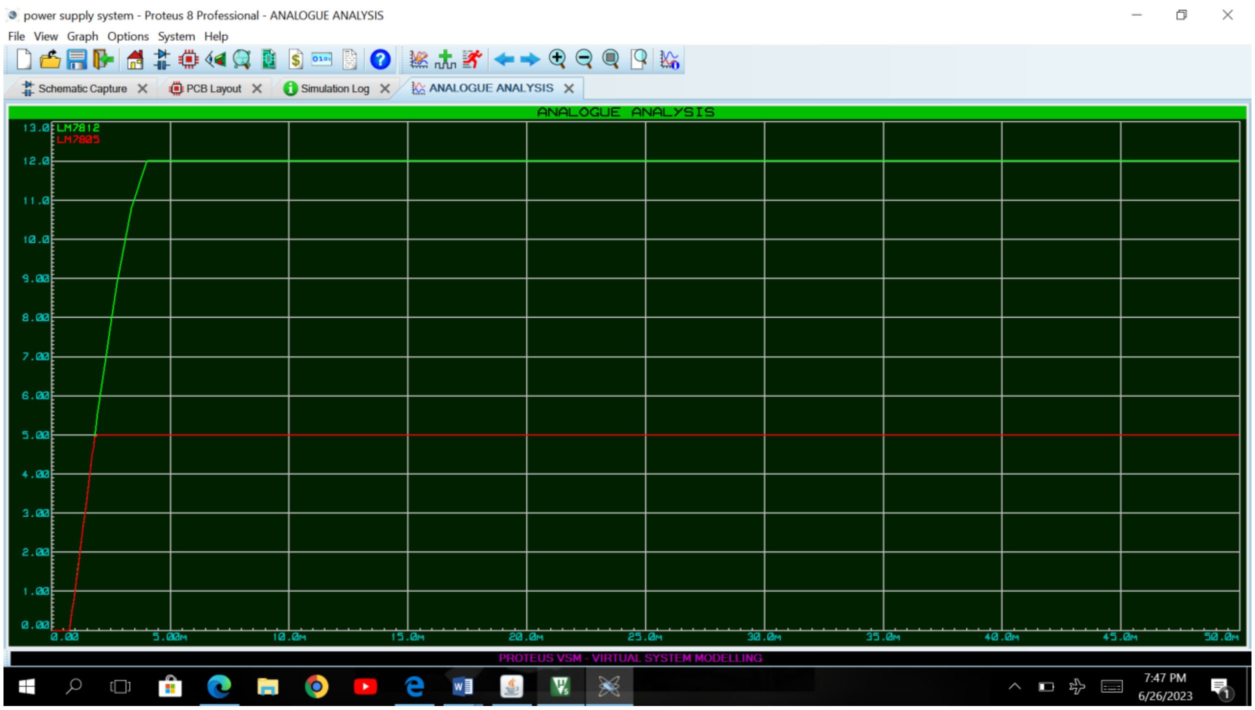


Figure 4.4.5 DC output voltage

3.3.2 Control system

To achieve my second objective which is to design a system that will be easily installed, configured, and controlled by users with minimal technical expertise, I made use of two components to monitor and control our system. Those components are Bluetooth HC-05 module and LM016L LCD for the Display. Both interfaced with the 8051 microcontroller.

i) LM016L LCD interfacing 8051 Microcontroller.

My system aims to be user-friendly. And one of the ways to make it so is by providing an interface where the user will be able to see what he’s constantly doing and directs him in his automation. Thus, the need of an LCD screen. Below is the circuit diagram for the interface between the LCD and the 8051. We first make use of a virtual terminal as input for our home automation system.

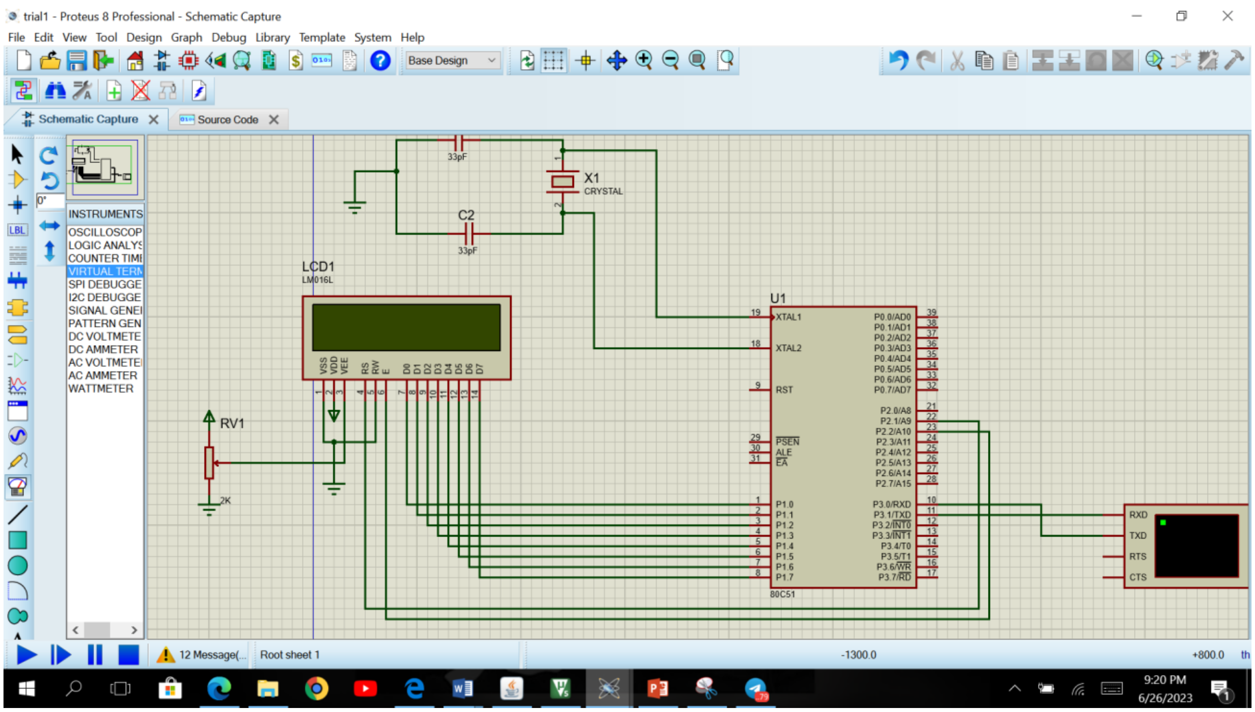


Figure 3.4.6 : LCD interface

For our system to display anything on the LCD we need to upload an assembly language or C code. The code below is a compiled assembly language code for our interface to print home automation on the LCD.

```
ORG 0000H

MOV A, #38H     ; use 2 lines and 5*7

ACALL com

MOV A, #0EH   ;cursor blinking off

ACALL com

MOV A, #80H   ;force cursor to first line

ACALL com

MOV A, #01H     ;clear screen

ACALL com
```

```
MOV DPTR, #STR
REV: MOV A, #00H
MOVC A, @A+DPTR
JZ FINISH
ACALL L_D
INC DPTR
SJMP REV
FINISH: SJMP FINISH

com: ACALL DEL_ROUTINE
MOV P1, A
CLR P2.1
SETB P2.2
CLR P2.2
RET

L_D: ACALL DEL_ROUTINE
MOV P1, A
SETB P2.1
SETB P2.2
CLR P2.2
RET

DEL_ROUTINE: MOV R0, #0FFH
L1: MOV R1, #0FFH
```

```
L2: DJNZ R1, L2

DJNZ R0, L1

RET

STR: DB 'HOME AUTOMATION',0

END
```

ii) Bluetooth Module

The choice for an interface to communicate with the microcontroller was very difficult. But considering the state of our present era, I was faced between the choice of a Bluetooth interface and WIFI interface. But considering the fact that our project aims for a developing country, so, there's still unstable internet connection and thus, my attention tend to a Bluetooth module. Below is the circuit diagram for the interface between the 8051 microcontroller and the Bluetooth HC-05 module.

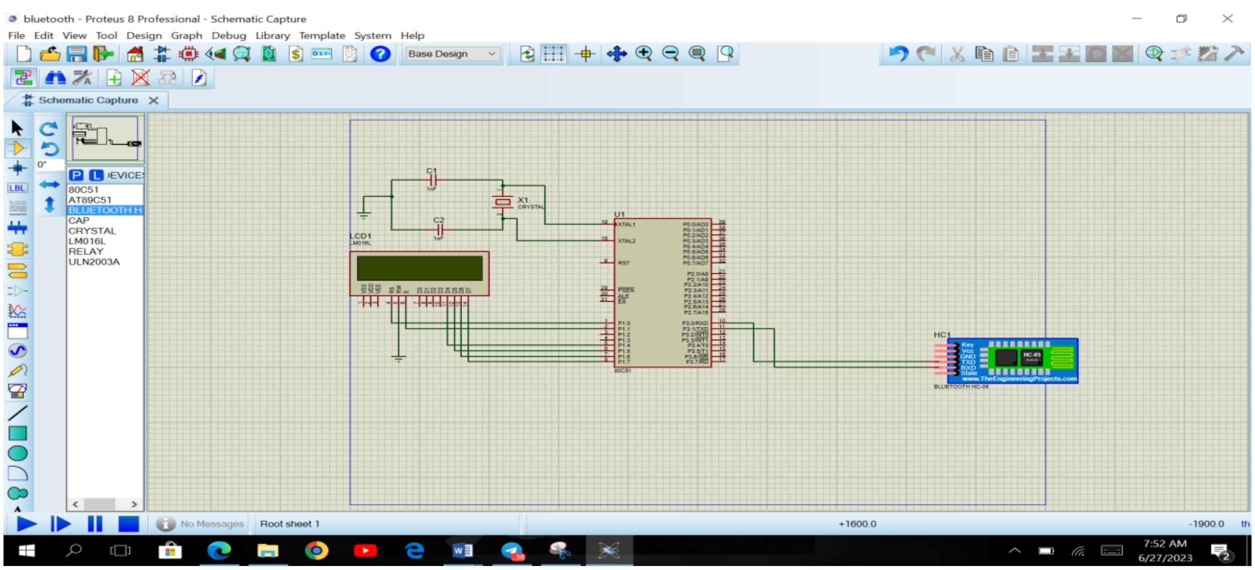


Figure 3.4.6 : Bluetooth interface

3.3.3 OUTPUT LOAD SYSTEM

Our output system is one of the most important part of the system since it is the part responsible to control our home appliances. The design of my output system was using the ULN2003A IC (relay module) and a 12V drive relay. The circuit below shows how the 8051 is interfaced with the ULN2003A to control a switch.

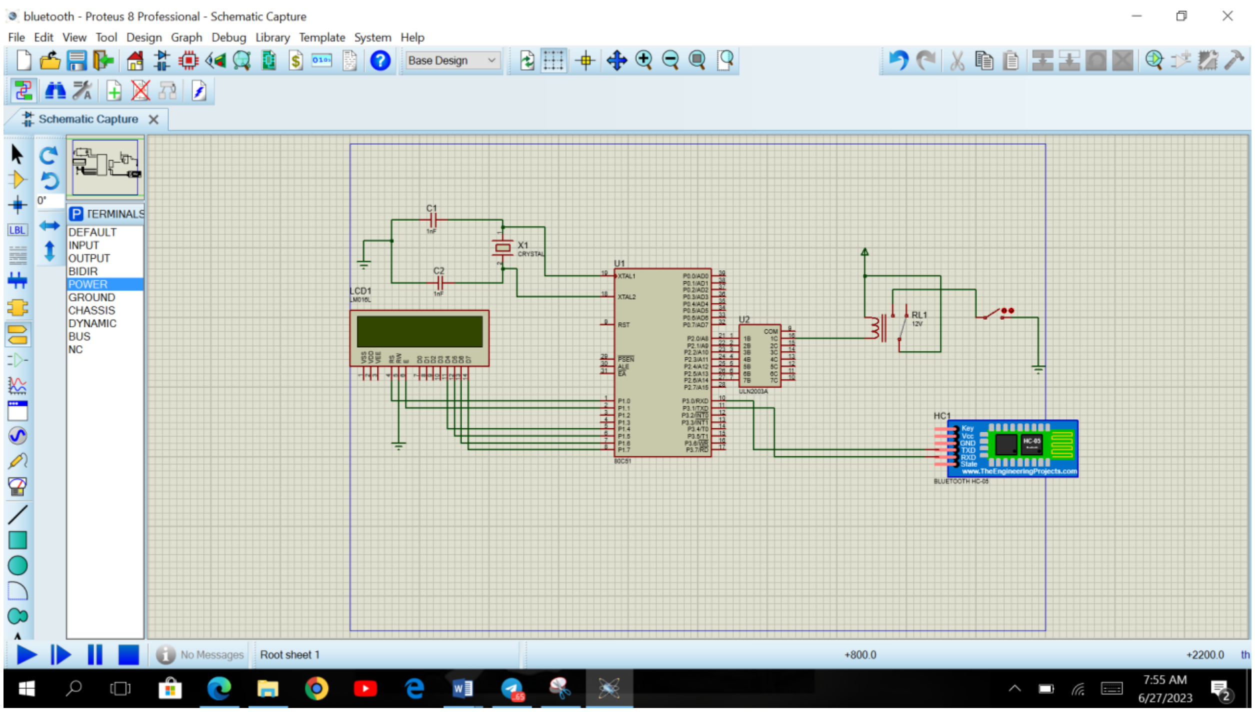
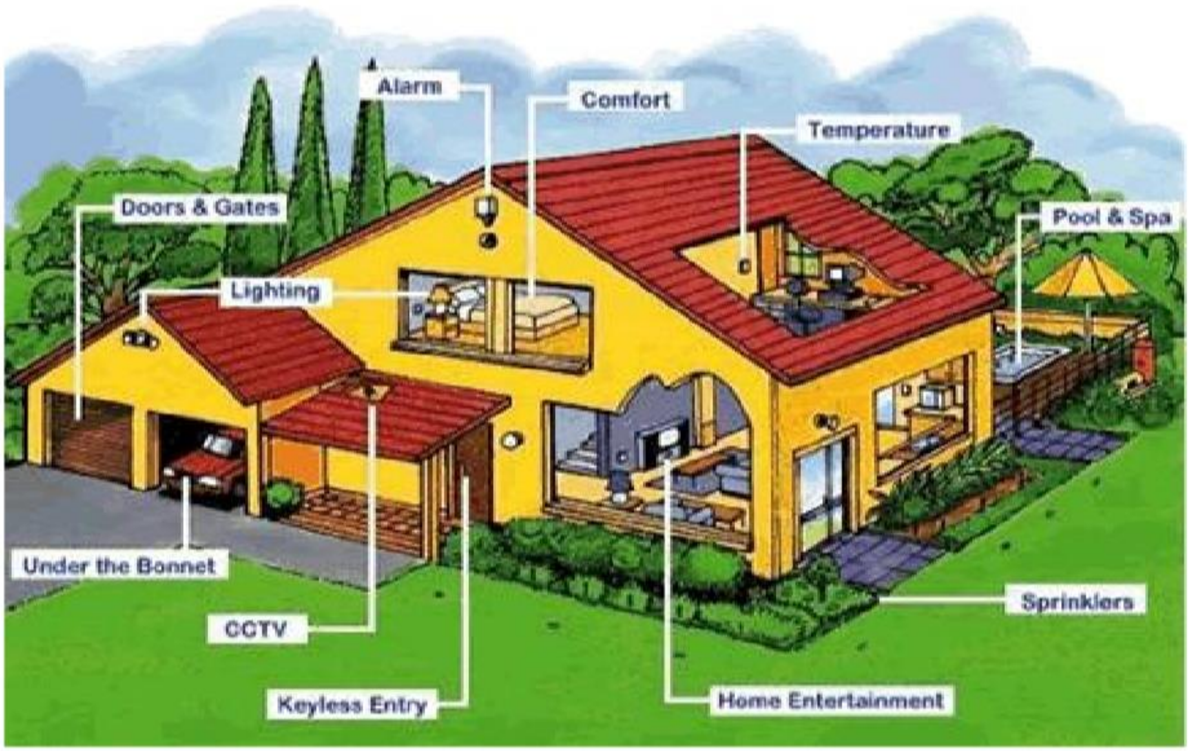


Figure 3.4.7: Load interface

3.3.4 CIRCUIT DIAGRAM

After the various interfacing, this section contains all those different sub-systems interfaced to give one system which is our home automation system.

Our idea in the beginning was to have a home automation system like in the figure below.



To design our system, we had to follow the following flowchart steps as guidelines:

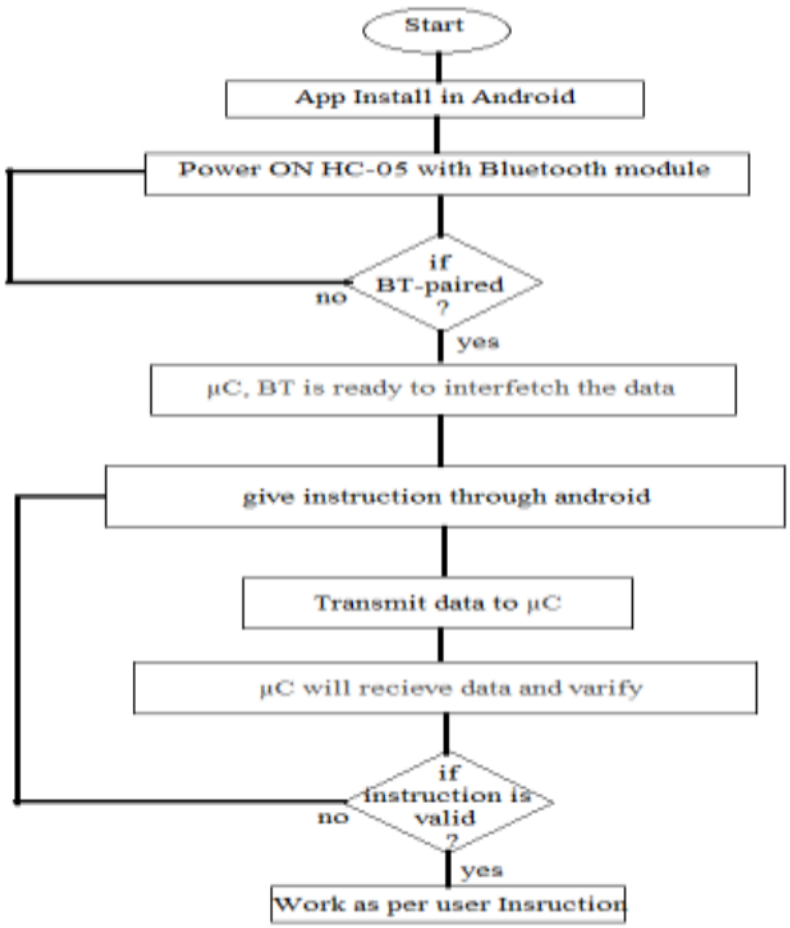
1. Start
2. Initialize the 8051 microcontroller and ICs
3. Set up the input/output pins for the microcontroller
4. Connect the HC-05 bluetooth module and LCD to the microcontroller
5. Connect the sensors and actuators (fan, gate, light, TV, alarm, doorbell ring) to the input/output pins
6. Pair the HC-05 bluetooth module with a smartphone or other device
7. Receive commands from the smartphone or other device via the HC-05 bluetooth module
8. Display the commands on the LCD
9. Analyze the received commands and determine the appropriate action to take
10. Control the actuators based on the analysis of the received commands

11. Display the status of the actuators on the LCD

12. Repeat steps 7-11 continuously

13. End

A brief flowchart is given in the diagram below;



We achieved our system by designing it using Proteus and its circuit is given in the following diagram.

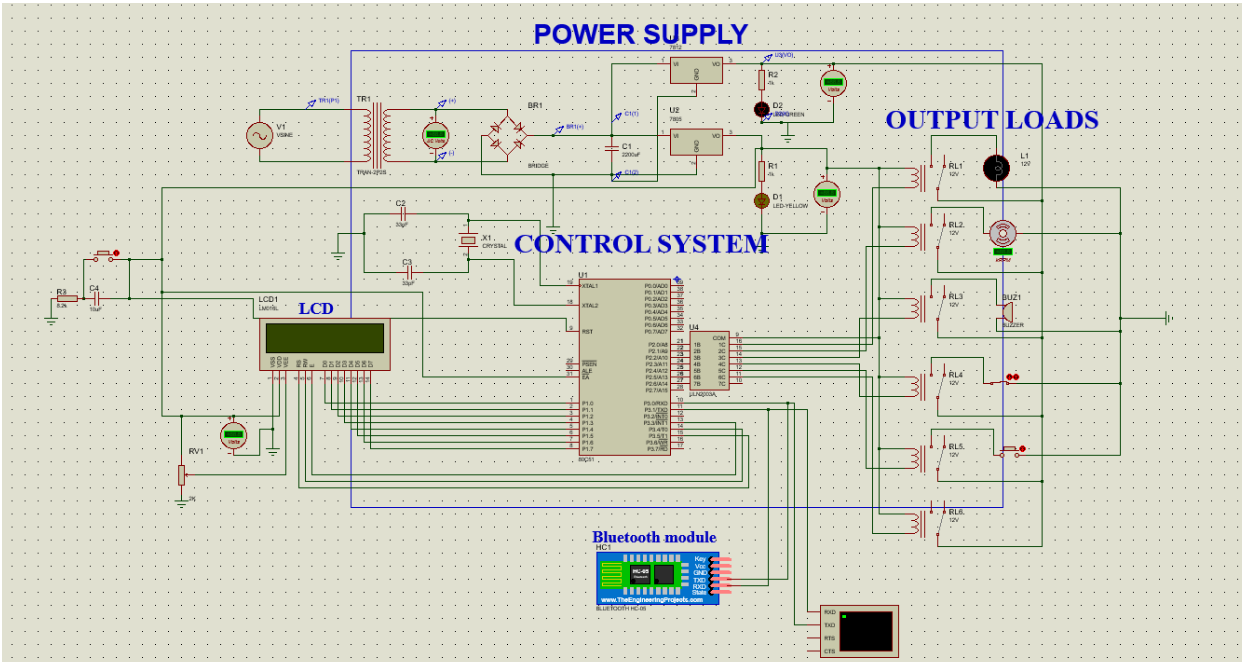


Figure 3.4.8: home automation Circuit diagram

3.3.5 System explanation

In the circuit diagram the microcontroller is connected to the home appliances using Bluetooth module and relays. Here, we employed HC-05 serial communication device which will transfer the characters to the microcontroller (8051). Now, 8051 decode all characters and generate special function and it will operate the relays. Here, we are used 5V 8-channel relay. Here, to create a circuit diagram of 8051 we internally connected reset circuit to create non-maskable interrupt on the 9th pin with 10µF capacitor and 8.2kΩ resister. Here, we used default crystal frequency 11.0592 MHz with two 10pF capacitor to the ground. And we are shorted the 31 pins to VCC to fetch the internal memory. We put reset button to externally reset the microcontroller. In the power supply we used 12-0-12 center tapped transformer to step down the voltage and bridge rectifier made of IN4001 diode and we put 1000µF capacitor to remove the ripples. The microcontroller requires 5V dc input so we used LM7805 IC regulator which gave a 5V fixed. And also used the LED for indication purpose. Then, the microcontroller output signal went to ULN2003A (npn-darlington transistor array) which drove the relays to control our respective home appliances.

3.3.6 Assembly Language Code

The use of assembly language code for home automation projects has become increasingly popular due to its ability to provide precise control over hardware components. In this project, we have designed a home automation system using the 8051 microcontroller and various integrated circuits (ICs). The system enables users to remotely control their home appliances using a user-friendly interface and provides real-time feedback on their status. The use of assembly language code has allowed us to optimize the system's performance, making it highly efficient and reliable. In this report, we will discuss the assembly language code used in the project and how it has contributed to the overall success of the home automation system.

Below is the assembly language code to interface the all the components with our system.

```
ORG 0000H

MOV P1, #00H ; Initialize port 1 for LCD
MOV P2, #00H ; Initialize port 2 for relay driver
MOV P3, #00H ; Initialize port 3 for Bluetooth module

CALL LCD_INIT ; Initialize LCD
CALL BLUETOOTH_INIT ; Initialize Bluetooth module

MAIN:
  CALL BLUETOOTH_RECEIVE ; Receive data from Bluetooth module
  CJNE A, #0FFH, LIGHT ; If data is 0xFF, go to LIGHT
  CJNE A, #0FEH, FAN ; If data is 0xFE, go to FAN
  CJNE A, #0FDH, GATE ; If data is 0xFD, go to GATE
  CJNE A, #0FCH, DOORBELL ; If data is 0xFC, go to DOORBELL
```

```
    CJNE A, #0FBH, ALARM ; If data is 0xFB, go to ALARM
    CJNE A, #0FAH, SECURITY ; If data is 0xFA, go to SECURITY
    JMP MAIN ; If data is not recognized, go back to MAIN

LIGHT:
    SETB P2.0 ; Turn on light
    CALL LCD_PRINT_LIGHT ; Print "Light On" on LCD
    JMP MAIN ; Go back to MAIN

FAN:
    SETB P2.1 ; Turn on fan
    CALL LCD_PRINT_FAN ; Print "Fan On" on LCD
    JMP MAIN ; Go back to MAIN

GATE:
    SETB P2.2 ; Open gate
    CALL LCD_PRINT_GATE ; Print "Gate Open" on LCD
    JMP MAIN ; Go back to MAIN

DOORBELL:
    SETB P2.3 ; Ring doorbell
    CALL LCD_PRINT_DOORBELL ; Print "Doorbell Ringing" on LCD
    JMP MAIN ; Go back to MAIN

ALARM:
    SETB P2.4 ; Activate alarm
    CALL LCD_PRINT_ALARM ; Print "Alarm Activated" on LCD
```

```
    JMP MAIN ; Go back to MAIN

SECURITY:
    SETB P2.5 ; Activate security system
    CALL LCD_PRINT_SECURITY ; Print "Security System Activated" on LCD
    JMP MAIN ; Go back to MAIN

LCD_INIT:
    MOV A, #38H ; Function set: 2 lines, 5x7 matrix
    ACALL LCD_COMMAND ; Send command to LCD
    MOV A, #0FH ; Display on, cursor blinking
    ACALL LCD_COMMAND ; Send command to LCD
    MOV A, #01H ; Clear display
    ACALL LCD_COMMAND ; Send command to LCD
    MOV A, #06H ; Entry mode set: increment cursor, no shift
    ACALL LCD_COMMAND ; Send command to LCD
    RET

LCD_COMMAND:
    CLR P1.0 ; RS = 0
    MOV P1.1, A ; Send command to LCD
    SETB P1.2 ; Enable pulse
    CLR P1.2 ; Clear enable pulse
    ACALL DELAY ; Delay for LCD to process command
    RET

LCD_PRINT_LIGHT:
```

```
    MOV A, #80H ; Set cursor to first line, first position
    ACALL LCD_COMMAND ; Send command to LCD
    MOV A, #"Light On" ; Send data to LCD
    ACALL LCD_DATA ; Send data to LCD
    RET

LCD_PRINT_FAN:
    MOV A, #80H ; Set cursor to first line, first position
    ACALL LCD_COMMAND ; Send command to LCD
    MOV A, #"Fan On" ; Send data to LCD
    ACALL LCD_DATA ; Send data to LCD
    RET

LCD_PRINT_GATE:
    MOV A, #80H ; Set cursor to first line, first position
    ACALL LCD_COMMAND ; Send command to LCD
    MOV A, #"Gate Open" ; Send data to LCD
    ACALL LCD_DATA ; Send data to LCD
    RET

LCD_PRINT_DOORBELL:
    MOV A, #80H ; Set cursor to first line, first position
    ACALL LCD_COMMAND ; Send command to LCD
    MOV A, #"Doorbell Ringing" ; Send data to LCD
    ACALL LCD_DATA ; Send data to LCD
    RET
```

```
LCD_PRINT_ALARM:
    MOV A, #80H ; Set cursor to first line, first position
    ACALL LCD_COMMAND ; Send command to LCD
    MOV A, #"Alarm Activated" ; Send data to LCD
    ACALL LCD_DATA ; Send data to LCD
    RET

LCD_PRINT_SECURITY:
    MOV A, #80H ; Set cursor to first line, first position
    ACALL LCD_COMMAND ; Send command to LCD
    MOV A, #"Security System Activated" ; Send data to LCD
    ACALL LCD_DATA ; Send data to LCD
    RET
LCD_DATA:
    SETB P1.0 ; RS = 1
    MOV P1.1, A ; Send data to LCD
 SETB P1.2 ; Enable pulse
  CLR P1.2 ; Clear enable pulse
  ACALL DELAY ; Delay for LCD to process data
    RET

BLUETOOTH_INIT:
    MOV SCON, #50H ; Set UART mode 1, enable receiver
    MOV TMOD, #20H ; Set timer 1 mode 2, 8-bit auto-reload
    MOV TH1, #-3 ; Set baud rate to 9600
    SETB TR1 ; Start timer 1
    RET
```

```
BLUETOOTH_RECEIVE:
    JNB RI, $ ; Wait for data to be received
    MOV A, SBUF ; Read received data
    CLR RI ; Clear receive interrupt flag
    RET
DELAY:
    MOV R7, #255
   DELAY1:
   MOV R6, #255
   DELAY2:
    DJNZ R6, DELAY2
   DJNZ R7, DELAY1
    RET

END
```

This program initializes the ports for the LCD, Bluetooth module, and relay driver, and then enters a loop to receive data from the Bluetooth module. Depending on the received data, it turns on/off the corresponding relay and prints a message on the LCD. The program also includes subroutines to initialize the LCD and Bluetooth module, send commands and data to the LCD, and receive data from the Bluetooth module. Finally, it includes a delay subroutine to provide a delay between LCD commands and data.

3.4 Conclusion

This chapter was concerned with the selection of components needed to build effectively our home automation system, designing our system and, implementing it on proteus and programming it using assembly language programming. We took into consideration the following while selecting our components.

- A brief overview
- its advantages and disadvantages
- functions and implementation
- compatibility with our 8051 microcontroller

CHAPTER 4: Conclusion and recommendation

4.1 CONCLUSION

to begin with, Designing and control of A Bluetooth based home automation system using 8051 microcontroller and ICs was of a truth a fascinating task to undergo as first time long essay. This project presented me a new way of understanding of electronics as it offered an interfaced which covered all what have been taught through my undergraduate curriculum. This project covered topics like basic electronics components, Integrated circuits, power electronics, communication systems, Networking, microcontrollers and programming which were all core courses for an undergraduate degree as applied physics students.

On the other hand, the system offers a convenient and efficient way to control various home appliances remotely, providing users with greater flexibility and ease of use. The project has shown that with the right tools and knowledge, it is possible to develop innovative solutions that can improve the quality of life for people in developing countries. As technology continues to advance, it is important to continue exploring new ways to leverage it for the benefit of society.

Overall, the home automation system using the 8051 microcontroller and interfacing Bluetooth module is a step in the right direction towards achieving this goal.

4.2 RECOMMENDATION

The main idea behind science is evolution. After our small research, I noticed that there are still many ways to improve my system. The following points will summarize what came into my mind.

- Including sensors

The project we have undertaken it can be taken as a greater level to show the capabilities of the system it can cover temperature updates, weather forecasting, system synchronization, etc.

- Complete home automation

The project itself can be modified to achieve a complete Home Automation System which can create a platform for the user to interface between himself and his household.

- Interface with wifi module

In future, the system will be more compact and handy with combining the microcontroller and WIFI module.

- Power efficiency

The electric failure shall not be taken place because hardware will be self-contained. This appliance will have its own power bank and charging system.

- Computer interfacing

The computer system will also be interfaced to record and process data base.

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